Lecture Notes · Linear Algebra · Session 1 — Fields and linear systems

These notes contain **all** the material of the session: what is explained in the videos plus the precise definitions, the complete proofs and the full examples that the videos only sketch. Exercises and a Going further section at the end.

1.1 Fields: the stage of linear algebra

Throughout the course we have computed in several "numerical worlds": $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and the $\mathbb{Z}/p\mathbb{Z}$ of modular arithmetic. All of them share one decisive property: in them one can **add, subtract, multiply and divide** (except by zero) without leaving the set. Linear algebra does not need to know *which* of these worlds we work in; it only needs that property. It pays, then, to isolate it.

Definition 4.1.1 (field). A **field** is a set K with two **internal operations**

$$+ : K \times K \to K, \qquad \cdot : K \times K \to K$$

(that is, the sum and the product of two elements of K land back in K) satisfying:

1. *(Addition)* For all $a, b, c \in K$:
   - associative: $(a+b)+c = a+(b+c)$;
   - commutative: $a+b = b+a$;
   - there is an additive **identity** $0 \in K$ with $a+0=a$;
   - every a has an **opposite** $-a$ with $a+(-a)=0$.
2. *(Multiplication)* For all $a, b, c \in K$:
   - associative: $(ab)c = a(bc)$;
   - commutative: $ab = ba$;
   - there is a multiplicative **identity** $1 \in K$, $1 \neq 0$, with $1 \cdot a = a$;
   - every $a \neq 0$ has an **inverse** a^{-1} with $a \cdot a^{-1} = 1$.
3. *(Distributivity)* $a(b+c) = ab+ac$.

The piece that sets a field apart from poorer structures is the last item of point 2: **every nonzero element has a multiplicative inverse**. In a field *one can divide*: $\frac{b}{a} := a^{-1}b$ for $a \neq 0$.

In the language of structures: $(K, +)$ and $(K \setminus \{0\}, \cdot)$ are *abelian groups*. We will use this name only as a convenient label.

Example 4.1.2.

1. $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are fields with their usual operations.
2. $\mathbb{Z}/p\mathbb{Z}$ with p **prime** is a field (finite, with p elements): every nonzero element is invertible modulo p. For instance, in $\mathbb{Z}/5\mathbb{Z}$ the inverses are $1^{-1}=1$, $2^{-1}=3$, $3^{-1}=2$ and $4^{-1}=4$ (check it: $2 \cdot 3 = 6 = 1$ and $4 \cdot 4 = 16 = 1$ modulo 5).

> 3. $\mathbb{Z}$ is **not** a field: 2 has no integer inverse. (Note that in $\mathbb{Z}$ there are no zero divisors; failing to be a field and having zero divisors are different defects.)
> 4. $\mathbb{Z}/6\mathbb{Z}$ is **not** a field: $2 \cdot 3 = 0$ with $2, 3 \neq 0$ (they are **zero divisors**) and, moreover, 2 is not invertible.

Remark 4.1.3 (zero divisor vs. non-invertible). These are two different properties: *zero divisor* speaks of products that vanish; *non-invertible*, of the impossibility of dividing. In $\mathbb{Z}$ there are non-invertible elements without there being any zero divisors. What happens in $\mathbb{Z}/n\mathbb{Z}$ is special: there, for nonzero elements, **being a zero divisor is equivalent to not being invertible**; that equivalence is proved in the Modular arithmetic module and here we use it only as context.

Proposition 4.1.4 (uniqueness of identities and inverses). In a field K: 1. The additive identity 0 and the multiplicative identity 1 are unique. 2. The opposite $-a$ of each a is unique, and the inverse a^{-1} of each $a \neq 0$ is unique.

Proof. (1) If 0 and $0'$ are additive identities, then $0 = 0 + 0' = 0'$, where the first equality uses that $0'$ is an identity and the second that 0 is. The same argument with 1 and $1'$: $1 = 1 \cdot 1' = 1'$.

(2) If b and c are opposites of a, then

$$b = b + 0 = b + (a + c) = (b + a) + c = 0 + c = c,$$

using only associativity, commutativity and the definition of opposite. Analogously, if b and c are inverses of $a \neq 0$:

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c. \qquad \blacksquare$$

Note that the uniqueness proof for the inverse is *identical in form* to the one for the opposite: it only uses associativity, the identity and the defining property. This pattern of argument will reappear (e.g., uniqueness of the inverse matrix, Session 2).

Course convention. Unless stated otherwise, we work over $K = \mathbb{R}$. But we will state the results as *"if K is a field…"*: this way they also hold for $\mathbb{C}$, $\mathbb{Q}$ and $\mathbb{Z}/p\mathbb{Z}$ at no extra cost.

Why a field? Solving linear equations is, ultimately, **isolating the unknown**, and isolating means **dividing by coefficients**: from $ax = b$ with $a \neq 0$ we pass to $x = a^{-1}b$. Without inverses there is no solving for x. We will see this principle *in action* inside Gaussian elimination (§1.3), pointing out the exact spot where it is used.

1.2 Systems of linear equations

Definition 4.1.5. A **linear equation** in the unknowns $x_1, \dots, x_n$ over a field K is an expression

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b, \qquad a_i, b \in K.$$

A **linear system** of m equations and n unknowns is a collection of m equations of this kind. A **solution** is an n-tuple $(s_1, \dots, s_n) \in K^n$ —the Cartesian product of n copies of K— that satisfies all the equations simultaneously. The **solution set** is the subset of K^n formed by all the solutions.

Geometric interpretation. In $\mathbb{R}^2$, each linear equation (with some nonzero coefficient) describes a **line**; in $\mathbb{R}^3$, a **plane**. The solution set of the system is the **intersection** of those objects. The picture scales to any dimension: in $\mathbb{R}^n$ each equation defines a **hyperplane**, and solving the system is still intersecting hyperplanes — going up in dimension is, in linear algebra, almost free. The hypothesis "*with some nonzero coefficient*" matters: if all the a_i vanish, the equation becomes $0 = b$, which **does not describe a line** — it is the **empty set** if $b \neq 0$ and **all** of $\mathbb{R}^2$ if $b = 0$.

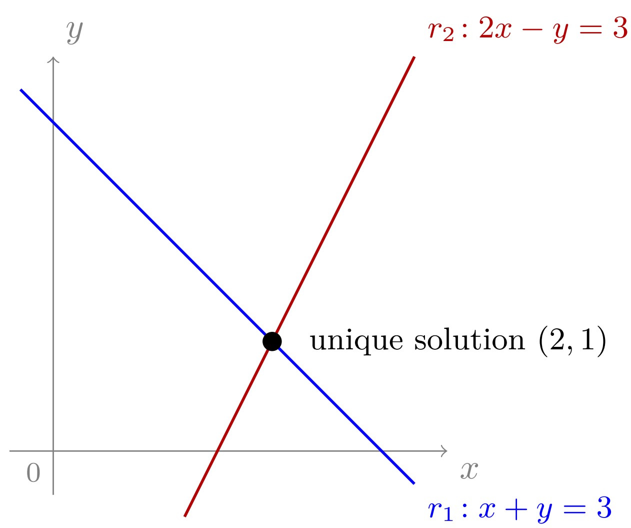

Figure 1: A 2×2 system consistent with a unique solution: each equation is a line and the solution is their single crossing point, here $(2, 1)$.

Definition 4.1.6 (classification). A system is:

- **consistent with a unique solution** if it has exactly one solution;

- **consistent with infinitely many solutions** if it has more than one (we shall see: infinitely many, if K is infinite);
- **inconsistent** if it has none.

In $\mathbb{R}^2$, the three cases correspond to lines meeting at a point, coincident lines, and distinct parallel lines. The rigorous justification of this trichotomy (the Rouché–Capelli criterion, known in Spanish textbooks as the Rouché–Frobenius theorem) will arrive in Session 4; in this session we will read it directly off Gaussian elimination.

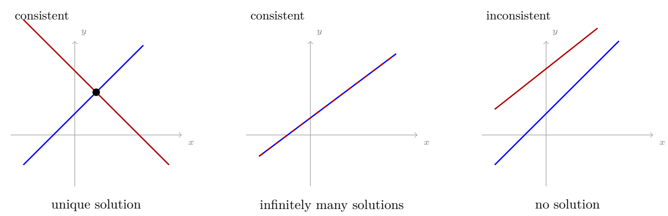

Figure 2: The three cases of a 2×2 system, geometrically: lines that cross (unique solution), coincident lines (infinitely many) or parallel lines (no solution).

1.3 Gaussian elimination: which transformations are legitimate

The strategy: transform the system into another one **with the same solution set** but of simpler shape (*echelon form*), from which the solutions can be read off by substitution. The crucial question is not *how* to transform, but *which transformations are legitimate*: which ones do not alter the solutions.

Definition 4.1.7 (elementary operations). On the equations (rows) of a system:

- **(T1)** swap two equations: $E_i \leftrightarrow E_j$;
- **(T2)** multiply an equation by $\lambda \in K$, $\lambda \neq 0$: $E_i \to \lambda E_i$;
- **(T3)** add to an equation a multiple of another: $E_i \to E_i + \mu E_j$ ($\mu \in K$, $j \neq i$).

Two systems are **equivalent** if one is obtained from the other by a chain of elementary operations.

Both restrictions are necessary. In (T2) we require $\lambda \neq 0$: with $\lambda = 0$ the operation turns E_i into $0 = 0$, an equation is lost and the solution set **may grow** (the operation stops being reversible: there is no 0^{-1}). In (T3) we require $j \neq i$: the condition prevents altering the very equation used as support — with $j = i$ the operation would be $E_i \to (1 + \mu)E_i$, which for $\mu = -1$ gives $0 = 0$

again and cannot be undone either. These are exactly the two hypotheses that the proof of Theorem 4.1.8 spends.

A concrete counterexample (for $\lambda = 0$). The system

$$\begin{cases} x + y = 1 \\ x - y = 3 \end{cases}$$

has the unique solution $(2, -1)$. Applying to E_2 the forbidden operation $E_2 \to 0 \cdot E_2$ leaves

$$\begin{cases} x + y = 1 \\ \qquad 0 = 0, \end{cases}$$

whose solution set is the **whole** line $x + y = 1$: solutions have appeared out of nowhere. The proof of the theorem fails exactly where it should, in the reverse inclusion: undoing the operation would require multiplying by 0^{-1}, which exists in no field.

> **Mind the moral.** A row $0 = 0$ **obtained legally** during elimination is not an error: it means one equation was redundant (see Example 4.1.16). What is illegitimate is to *manufacture* it with a non-reversible operation, because then information has been thrown away. The difference is made by reversibility, not by the look of the row.

Theorem 4.1.8 (elementary operations preserve the solutions). If a system S' is obtained from S by applying one elementary operation, then S and S' have **the same solution set**.

Proof. We prove the two inclusions.

($\subseteq$) Every solution of S is a solution of S'. Let $s = (s_1, \dots, s_n)$ be a solution of S; we check that it satisfies the equations of S', which coincide with those of S except for the modified one.

- **(T1)** does not change the equations, only their order: trivial.
- **(T2)** the new equation is λE_i. If s satisfies E_i, that is $\sum_k a_k s_k = b$, multiplying both sides by λ: $\sum_k (\lambda a_k) s_k = \lambda b$. Hence s satisfies λE_i.
- **(T3)** the new equation is $E_i + \mu E_j$. If s satisfies E_i ($\sum_k a_k s_k = b$) and E_j ($\sum_k c_k s_k = d$), adding the second multiplied by μ to the first: $\sum_k (a_k + \mu c_k) s_k = b + \mu d$. Hence s satisfies $E_i + \mu E_j$.

($\supseteq$) Every solution of S' is a solution of S. It suffices to observe that **each operation is reversible by another elementary operation of the same type**:

- (T1) is undone by the same swap;
- (T2), $E_i \to \lambda E_i$, is undone by $E_i \to \lambda^{-1} E_i$ — **and here is where we use that K is a field**: we need λ^{-1} to exist, and it exists because $\lambda \neq 0$;
- (T3), $E_i \to E_i + \mu E_j$, is undone by $E_i \to E_i - \mu E_j$ (the equation E_j was not altered).

Thus S is obtained from S' by an elementary operation, and by the inclusion already proved every solution of S' is a solution of S. ■

Remark 4.1.9. The theorem is the *user license* of Gaussian elimination: we may chain operations with complete freedom, knowing that the solution set is untouched. Note the double role of the field: the reversibility of (T2) uses λ^{-1}, and back substitution (§1.4) will divide by pivots.

1.4 The table and the staircase

While one operates, the unknowns take no part: they only record which coefficient combines with which, and that information is already carried by the **position**. So we stop writing them: first we fix the table that remains (§ *The augmented matrix*), and on it we say **where** we are transforming to (§ *Echelon forms*).

The augmented matrix

Definition 4.1.10. Given a system of m equations and n unknowns with coefficients a_{ij} and right-hand sides b_i, its **coefficient matrix** A is the $m \times n$ table with entry a_{ij} in row i, column j; its **augmented matrix** is

$$[A \mid b] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right].$$

Index convention (in force for the whole module). In a_{ij} the **first** index is the **row** and the second the **column**. With it, the translation is complete:

$$\text{equation } i \longleftrightarrow \text{row } i, \qquad \text{unknown } j \longleftrightarrow \text{column } j,$$

and the elementary operations (T1), (T2), (T3) on equations are **exactly** the same operations on the **rows** of $[A \mid b]$. There is nothing new to prove: Theorem 4.1.8 reads verbatim on the table. For instance,

$$\begin{cases} 2x + 3y = 7 \\ x - y = 1 \end{cases} \quad \longleftrightarrow \quad \left[\begin{array}{cc|c} 2 & 3 & 7 \\ 1 & -1 & 1 \end{array}\right], \qquad E_2 \to E_2 - \tfrac{1}{2}E_1 \;=\; F_2 \to F_2 - \tfrac{1}{2}F_1.$$

For now the matrix is mere **notation**; in Session 2 it will come to life: from it will emerge operations (sum, product, inverse) with meaning.

Echelon forms

Definition 4.1.11 (echelon form). A matrix is in (row) **echelon form** if: 1. the zero rows (if any) are at the bottom; 2. the first nonzero coefficient of each nonzero row —its **pivot**— lies strictly to the right of the pivot of the previous row.

Condition 2 draws a staircase with nothing but zeros below it. Note that a step may **skip several columns** (that is legal, and those pivot-free columns are the ones that will give free unknowns); what it may not do is go back or repeat a column:

$$\underbrace{\left[\begin{array}{ccccc|c} p & * & * & * & * & * \\ 0 & 0 & p & * & * & * \\ 0 & 0 & 0 & 0 & p & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]}_{\text{echelon form}} \qquad \underbrace{\left[\begin{array}{ccccc|c} p & * & * & * & * & * \\ 0 & 0 & p & * & * & * \\ 0 & p & * & * & * & * \end{array}\right]}_{\text{not: the pivot goes back}}$$

Pivots are looked for in A, in front of the bar. A row with zeros throughout the A part and a nonzero entry behind the bar has no pivot: it is the equation $0 = c$, and it will decide the outcome of the system (see below).

Definition 4.1.12 (reduced echelon form). A matrix in echelon form is in **reduced echelon form** if in addition: 3. all pivots equal 1; 4. each pivot is the only nonzero element of its column.

> There are variants of these definitions in the literature (some texts require pivots = 1 already in the echelon form). The essential thing is the **staircase of pivots** (Def. 4.1.11); the reduced form (4.1.12) is the "maximally clean" version. To solve systems the echelon form suffices; the reduced form is convenient for theoretical results.

Theorem 4.1.13 (existence). Every matrix can be brought to echelon form by elementary operations (and to the reduced form, if desired).

Proof (algorithmic, by induction on the number of columns). If the first column is zero, ignore it and apply induction to the rest. Otherwise, there is a nonzero coefficient in it; with (T1) we bring it to the first row (it will be the first pivot, call it p) and with (T3) we kill all the coefficients below it: from the row with coefficient c we subtract $\frac{c}{p} = c\,p^{-1}$ times the first. **Here again we use that** K **is a field**: the "suitable multiple" is $c\,p^{-1}$, and it needs the inverse of the pivot — this is, once more, the central point of the session. The submatrix left after removing the first row and the first column has fewer columns; by induction it can be brought to echelon form with operations that touch neither the first row nor reintroduce anything below the pivot (they operate to its right). The result is in echelon form. For the reduced form: with (T2) each pivot is normalized to 1 (multiplying by p^{-1}) and with (T3), working from the bottom up, the coefficients *above* each pivot are killed. ■

Remark 4.1.14 (uniqueness). The echelon form of a matrix is **not** unique (different pivot choices give different echelon forms). The **reduced form is unique** — we state this without proof; it can be found in Hefferon, *Linear Algebra* (ch. One, section III). What does not depend on the choices (number and position of pivot columns) will underpin the **rank** (Session 3).

Back substitution and reading off the three cases

With the system in echelon form, one solves **from the bottom up**. Take the last nonzero row, with pivot p_r in column j_r. By the staircase shape, to the left of j_r that row has only zeros, so its equation involves x_{j_r} and, at most, unknowns **further to the right**:

$$p_r x_{j_r} + \sum_{j>j_r} a_{rj} x_j = b_r \qquad \Longrightarrow \qquad x_{j_r} = p_r^{-1}\Big(b_r - \sum_{j>j_r} a_{rj} x_j\Big),$$

where dividing by the pivot is —once again— using that K is a field. One goes up a step and repeats: **each pivot solves for its own unknown**, in terms of those not yet fixed.

The degenerate rows and the pivots dictate the type of system:

- a row $0 = 0$ always holds: the equation was **redundant** and can be dropped — *by itself it implies nothing more*;
- a row $0 = c$ with $c \neq 0$ never holds: the system is **inconsistent**, no matter what the rest looks like.

If there are no rows $0 = c$, what decides between a unique solution and infinitely many **is not the presence of rows $0 = 0$, but the pivots**: if **every unknown** (column of A) has a pivot, the solution is **unique**; if some unknown is left **without a pivot**, that unknown is **free** —it takes any value and the others are solved for in terms of it— and there are **infinitely many** solutions (over an infinite field; the exact count for any K, finite ones included, is Corollary 4.4.5b of Session 4: $|K|^{n-r}$).

Remark 4.1.14b (frequent error). "There is a row $0 = 0$" does **not** imply that there are free unknowns: with more equations than unknowns an equation may be redundant (a row $0 = 0$) and there may still be a pivot in every column — unique solution. Redundancy speaks of the *rows*; freedom, of the *columns*. The precise detail —*parametrization*— is developed in Session 4.

Complete examples

Example 4.1.15 (consistent, unique solution). We solve over $\mathbb{R}$, with the system and its augmented matrix side by side:

$$\begin{cases} x + y + z = 6 \\ x + 2y + 3z = 14 \\ x + 4y + 9z = 36 \end{cases} \qquad \longleftrightarrow \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 4 & 9 & 36 \end{array}\right]$$

Step 1. Zeros below the first pivot with (T3): $F_2 \to F_2 - F_1$ and $F_3 \to F_3 - F_1$.
Step 2. Zeros below the second: $F_3 \to F_3 - 3F_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 4 & 9 & 36 \end{array}\right] \xrightarrow{\substack{F_2-F_1 \\ F_3-F_1}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 3 & 8 & 30 \end{array}\right] \xrightarrow{F_3-3F_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 2 & 6 \end{array}\right]$$

The matrix is in echelon form, with no degenerate rows and with three **pivots** (1, 1, 2) for the three columns of A: **consistent with a unique solution** — and this is asserted *before* computing any unknown.

Step 3 (back substitution). From $2z = 6$, dividing by the pivot 2 (the field!): $z = 3$. Substituting into $y + 2z = 8$: $y = 2$. And into the first: $x = 6 - 2 - 3 = 1$.

$$\text{Unique solution: } (x, y, z) = (1, 2, 3).$$

Check (always good practice, and in the **original** equations): $1 + 2 + 3 = 6$ ✓; $1 + 4 + 9 = 14$ ✓; $1 + 8 + 27 = 36$ ✓.

Note that all the work happened on the table, **without writing a single unknown**.

Example 4.1.16 (the other two outcomes, with the same system). We change only the third equation of the previous example to $x + 3y + 5z = c$ and apply **the same** operations:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 1 & 3 & 5 & c \end{array}\right] \xrightarrow[F_3 - F_1]{F_2 - F_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 2 & 4 & c-6 \end{array}\right] \xrightarrow{F_3 - 2F_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & c-22 \end{array}\right]$$

Case $c = 22$ (consistent, infinitely many solutions). The last row becomes $0 = 0$: the third equation was **redundant** —indeed $E_3 = 2E_2 - E_1$, and the elimination discovered it by itself—. There are pivots in the columns of x and y, but **not in that of** z: the unknown z is **free**. Setting $z = t$ with arbitrary $t \in \mathbb{R}$, back substitution gives $y = 8 - 2t$ and $x = 6 - (8 - 2t) - t = -2 + t$, that is

$$(x, y, z) = (-2 + t,\ 8 - 2t,\ t) = (-2, 8, 0) + t\,(1, -2, 1), \qquad t \in \mathbb{R}:$$

a **line** of solutions (the three planes meet in a line, not in a point). Check with $t = 0$: $(-2, 8, 0)$ ✓; with $t = 1$: $(-1, 6, 1)$ ✓.

Case $c = 23$ (inconsistent). The last row becomes $0 = 1$, which never holds: the solution set is $\emptyset$. The reading is transparent: the third equation is a consequence of the first two, which force it to equal 22; demanding 23 is asking for two incompatible things at once.

> In the case $c = 22$ a row $0 = 0$ and a free unknown appear together, and it is tempting to read one as the cause of the other. They are not: see Remark 4.1.14b.

Exercises

1. ★ Solve by Gaussian elimination, identifying at each step the elementary operation used, and classify:

(a) $\begin{cases} 2x + y = 5 \\ x - 3y = -1 \end{cases}$ (b) $\begin{cases} x + y + z = 1 \\ 2x + 2y + 2z = 2 \\ x - y = 0 \end{cases}$ (c) $\begin{cases} x + y = 1 \\ 2x + 2y = 3 \end{cases}$

2. ★ Discuss according to the parameter $a \in \mathbb{R}$: $\begin{cases} x + y = 2 \\ ax + y = 1 \end{cases}$. For which values is it consistent with a unique solution, with infinitely many, inconsistent?
3. ★ **(Structural counterfactual.)** Consider the equations $2x = 1$ and $2x = 4$.
   (0) Solve them first over $\mathbb{Z}$: $2x = 1$ has **no** integer solution, whereas $2x = 4$ has **exactly one** ($x = 2$). Over $\mathbb{Z}$ existence fails, but uniqueness does not.
   (a) Solve them over $\mathbb{Z}/6\mathbb{Z}$ **by trying all six possible values** of x. Check that $2x = 1$ has **no** solution and that $2x = 4$ has **two** ($x = 2$ and $x = 5$).
   (b) Solve them over $\mathbb{Z}/5\mathbb{Z}$ and check that each has **exactly one** solution.
   (c) Explain the contrast: what property does 2 have in $\mathbb{Z}/5\mathbb{Z}$ that it lacks in $\mathbb{Z}/6\mathbb{Z}$, and at which step of "solving" $x = 2^{-1}b$ is it used? Conclude: in a field, $ax = b$ with $a \neq 0$ **always has exactly one** solution; outside a field, both existence *and* uniqueness can fail. *(Careful: with systems of several equations, a bad pivot can sometimes be dodged by choosing another; that is why this counterfactual uses a single equation, where division by 2 is unavoidable.)*
4. ★ Prove that in a field $a \cdot 0 = 0$ for every a. *(Hint: $a \cdot 0 = a(0 + 0)$ and use distributivity with additive cancellation.)*
5. Prove that in a field there are no zero divisors: if $ab = 0$ then $a = 0$ or $b = 0$. *(Hint: if $a \neq 0$, multiply by a^{-1}.)* Conclude that $\mathbb{Z}/6\mathbb{Z}$ cannot be a field, without examining inverses one by one.
6. Write the proof of Theorem 4.1.8 for the operation (T2) with all quantifiers explicit ("for every solution s…").
7. Complete case (c) of exercise 1 by writing its augmented matrix and pointing out the degenerate row that betrays the inconsistency.
8. Give a 3×3 matrix that admits two different echelon forms, exhibiting the two chains of operations (this illustrates Remark 4.1.14).
9. ★ In Example 4.1.16, for which values of c is the system consistent? Check moreover that the third equation is $2E_2 - E_1$, and explain, without doing any new computation, why that forces $c = 22$.

Going further and references

Going further (finite fields). Besides $\mathbb{Z}/p\mathbb{Z}$ there are finite fields of size p^k for each prime p and each $k \geq 1$ (the *Galois fields* $\mathbb{F}_{p^k}$), and there are none of any other size. See Childs, *A Concrete Introduction to Higher Algebra*, or Aluffi, *Algebra: Chapter 0*, ch. V.

References for this session. Systems and Gaussian elimination: Hefferon, *Linear Algebra*, ch. One (free, with solutions); Strang, *Introduction to Linear Algebra*, ch. 1–2. Fields: Dorronsoro–Hernández, *Números, grupos y anillos*; Aluffi, ch. 0 and V. Uniqueness of the reduced form: Hefferon, One.III.

Lecture Notes · Linear Algebra · Session 2 — The algebra of matrices

All the material of the session, with the constructions in full: the matrix · vector product and its properties, the matrix product **derived** from composition (with the general proof), the elementary matrices, and the inverse with its properties proved. Exercises and a Going further section at the end.

2.1 The matrix · vector product: $Ax = b$

In Session 1 the matrix was pure notation. The first step towards giving it life: writing a whole system in a single expression.

Definition 4.2.1 (matrix · vector product). The product is an **operation** with its own domain and codomain,

$$K^{m\times n} \times K^n \longrightarrow K^m, \qquad (A, x) \longmapsto Ax,$$

where $K^{m\times n}$ denotes the $m \times n$ matrices over K: given A of size $m \times n$ and $x = (x_1, \dots, x_n) \in K^n$, the product Ax is the vector **in** K^m whose i-th component is

$$(Ax)_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{j=1}^{n} a_{ij}x_j.$$

It is worth pausing on the three sizes, since they are the module's first compatibility rule: the product is **defined only** when the number of **columns** of A equals the number of components of x; and the result **does not live in K^n, but in K^m** — a vector with one entry per unknown goes in, one with one entry per equation comes out.

The rule is not arbitrary: $(Ax)_i$ is *literally* the left-hand side of the i-th equation. With it, the system of Session 1 is written

$$Ax = b.$$

Definition 4.2.1b (sum of matrices and product by a scalar). For A, B of the **same** size $m \times n$ and $\lambda \in K$, the matrices $A + B$ and λA are the $m \times n$

matrices defined **entry by entry**:

$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (\lambda A)_{ij} = \lambda\, a_{ij}.$$

(Note the contrast with the product of matrices, which is ***not*** *entry by entry: we shall see it in §2.2.)* With these two operations, the $m \times n$ matrices form a vector space over K (Session 3, Example 4.3.2.3).

Proposition 4.2.2 (the product respects sums and scalars). For $u, v \in K^n$ and $\lambda \in K$:

$$A(u+v) = Au + Av, \qquad A(\lambda u) = \lambda(Au).$$

Proof. Component by component: $(A(u+v))_i = \sum_j a_{ij}(u_j + v_j) = \sum_j a_{ij}u_j + \sum_j a_{ij}v_j = (Au)_i + (Av)_i$, using the distributivity of K. Same with λ: $\sum_j a_{ij}(\lambda u_j) = \lambda \sum_j a_{ij}u_j$. ■

> This property, modest in appearance, will carry half the module: with it we will prove that the solutions of $Ax = 0$ form a subspace (Session 3), and it will be the model for the central notion of Session 5.

Proposition 4.2.3 (reading by columns). If $c_1, \dots, c_n \in K^m$ are the columns of A, then

$$Ax = x_1c_1 + x_2c_2 + \cdots + x_nc_n.$$

Proof. The i-th component of the right-hand side is $x_1a_{i1} + \cdots + x_na_{in}$, which is $(Ax)_i$. ■

> **Reading:** $Ax = b$ has a solution exactly when b can be manufactured by combining the columns of A. This observation will be the key to the consistency criterion (Session 4).

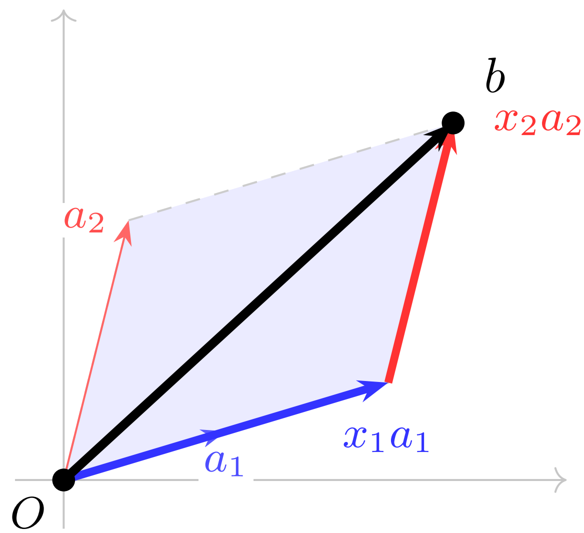

Figure 3: Reading by columns: $Ax = x_1a_1 + x_2a_2$ is a combination of the columns of A; the system $Ax = b$ is consistent when b lies in the span of the columns.

2.2 The matrix product, from composition

Motivation. If $y = Ax$ and afterwards $z = By$ (with compatible sizes: A of size $p \times n$, B of size $m \times p$), then $z = B(Ax)$. Is there a single matrix C with $z = Cx$ for all x? Let us compute it in the 2×2 case, with $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$:

$$z_1 = e\,y_1 + f\,y_2 = e(ax_1 + bx_2) + f(cx_1 + dx_2) = (ea + fc)\,x_1 + (eb + fd)\,x_2,$$
$$z_2 = g\,y_1 + h\,y_2 = g(ax_1 + bx_2) + h(cx_1 + dx_2) = (ga + hc)\,x_1 + (gb + hd)\,x_2.$$

The coefficients that appear are, in each position, *a row of B against a column of A*. The “row times column” rule is not postulated: composition **imposes** it.

Definition 4.2.4 (matrix product). For B of size $m \times p$ and A of size $p \times n$, the product BA is the $m \times n$ matrix with entries

$$(BA)_{ij} = \sum_{k=1}^{p} b_{ik}\,a_{kj} \qquad (\text{row } i \text{ of } B \cdot \text{column } j \text{ of } A).$$

Theorem 4.2.5. For every $x \in K^n$: $B(Ax) = (BA)\,x$.

Proof. The i-th component of the left-hand side:

$$(B(Ax))_i = \sum_{k=1}^{p} b_{ik}(Ax)_k = \sum_{k=1}^{p} b_{ik} \sum_{j=1}^{n} a_{kj} x_j$$

$$= \sum_{j=1}^{n} \left(\sum\nolimits_{k=1}^{p} b_{ik} a_{kj} \right) x_j = \sum_{j=1}^{n} (BA)_{ij}\, x_j = ((BA)x)_i\,,$$

where we have merely reordered finite sums (associativity and commutativity of K). ■

Lemma 4.2.6 (matrices are distinguished by their action). If M, N are $m \times n$ and $Mx = Nx$ for every $x \in K^n$, then $M = N$.

Proof. Taking $x = e_j$ (the vector with 1 in position j and 0 elsewhere), Me_j is the j-th column of M. Hence all the columns coincide. ■

Proposition 4.2.7 (properties of the product). 1. **Associative:** $(CB)A = C(BA)$ (compatible sizes). 2. **Distributive:** $B(A + A') = BA + BA'$ and $(B + B')A = BA + B'A$. 3. **Identity:** $I_m B = B = BI_p$, where I_k is the $k \times k$ **identity** matrix (ones on the diagonal, zeros elsewhere). In particular, $I_n x = x$ for every $x \in K^n$ (immediate from Definition 4.2.1: row i of I_n has a single 1, in position i). 4. **NOT commutative** in general.

Proof. (1) By Theorem 4.2.5, for every x: $((CB)A)x = (CB)(Ax) = C(B(Ax)) = C((BA)x) = (C(BA))x$; conclude with Lemma 4.2.6. *(Associativity is, at bottom, that of composing processes.)* (2) Direct computation with sums (exercise 5). (3) Immediate from the definition. (4) Counterexample:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},\ B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}: \quad AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = BA. \ ■$$

The transpose *(notes-only material)*

Definition 4.2.8. The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix A^T with $(A^\mathsf{T})_{ij} = a_{ji}$ (rows and columns interchanged).

Basic properties: $(A^\mathsf{T})^\mathsf{T} = A$, $(A + B)^\mathsf{T} = A^\mathsf{T} + B^\mathsf{T}$, and the important

$$(BA)^\mathsf{T} = A^\mathsf{T} B^\mathsf{T} \quad \text{(the order reverses!)},$$

whose verification with sums is exercise 6. The transpose will be used in Sessions 4 (one-sided inverses) and 6 ($\det A^\mathsf{T} = \det A$).

2.3 The Gaussian operations are products: elementary matrices

Definition 4.2.9 (elementary matrices). An **elementary matrix** is the identity I_m after applying **one** elementary row operation to it. There are three types, corresponding to (T1), (T2), (T3) of Session 1. In size $m \times m$ they are written with the identity in the background and **only what the operation changes**. Swapping rows i and j:

$$E_{(\mathrm{T1}),ij} = \begin{pmatrix} 1 & & & & \\ & 0 & \cdots & 1 & \\ & \vdots & \ddots & \vdots & \\ & 1 & \cdots & 0 & \\ & & & & 1 \end{pmatrix} \begin{array}{l} \\ \leftarrow i \\ \\ \leftarrow j \\ \\ \end{array}$$

multiplying row i by $\lambda \neq 0$:

$$E_{(\mathrm{T2}),i,\lambda} = \begin{pmatrix} 1 & & & \\ & \lambda & & \\ & & \ddots & \\ & & & 1 \end{pmatrix} \begin{array}{l} \\ \leftarrow i \\ \\ \\ \end{array}$$

and adding to row i a multiple μ of row j ($i \neq j$), where Δ_{ij} has a single 1 in position (i, j):

$$E_{(\mathrm{T3}),ij,\mu} = I + \mu\,\Delta_{ij} = \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & \vdots & \ddots & & \\ & \mu & \cdots & 1 & \\ & & & & 1 \end{pmatrix} \begin{array}{l} \\ \leftarrow j \\ \\ \leftarrow i \\ \\ \end{array}$$

In size 2×2, which is how they show up in the exercises:

$$E_{(\mathrm{T1})} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad E_{(\mathrm{T2}),\lambda} = \begin{pmatrix} 1 & 0 \\ 0 & \lambda \end{pmatrix} \ (\lambda \neq 0), \qquad E_{(\mathrm{T3}),\mu} = \begin{pmatrix} 1 & 0 \\ \mu & 1 \end{pmatrix}.$$

Lemma 4.2.10. If E is the elementary matrix of an operation, then EA is the result of applying that operation to A.

Proof (type (T3); the others, exercise 7). Let $E = I + \mu\,\Delta_{ij}$ where Δ_{ij} has a single 1 in position (i, j), $i \neq j$. Row r of EA is $\sum_k E_{rk}$ (row k of A). For $r \neq i$, $E_{rk} = \delta_{rk}$, so the row is left intact. For $r = i$, $E_{ik} = \delta_{ik} + \mu\,\delta_{jk}$, so row i of EA is (row i) $+\,\mu$ (row j): exactly the operation (T3). ■

Corollary 4.2.11. Reducing to echelon form by Gaussian elimination is multiplying on the left by a chain of elementary matrices: if A' results from A after k operations, $A' = E_k \cdots E_2 E_1 A$.

2.4 The inverse matrix

Definition 4.2.12. A **square** matrix A ($n \times n$) is **invertible** if there exists B ($n \times n$) with

$$AB = BA = I_n.$$

Such a B is called the **inverse** of A and written A^{-1}.

Proposition 4.2.13 (uniqueness). If A is invertible, its inverse is unique.

Proof. If B and C satisfy the definition, $B = BI = B(AC) = (BA)C = IC = C$, using only associativity. *(The same pattern as the uniqueness of the inverse in a field, Proposition 4.1.4.)* ■

Remark 4.2.14 (one side suffices — for square matrices). In the definition we require both equalities, but for **square** matrices one suffices: if $BA = I$ then automatically $AB = I$. This is **not evident** and will be proved with the machinery of rank (notes of Session 3, Corollary 4.3.24). For **non-square** matrices the story changes: an "inverse" may exist on **one side only** (Session 4, §4.4).

Proposition 4.2.15. 1. If A, B are invertible, AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$ (order reversed!). 2. Every elementary matrix is invertible, and its inverse is an elementary matrix of the same type.

Proof. (1) $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and likewise on the other side. (2) Each elementary operation is undone by another of the same type (Theorem 4.1.8): the matrix of the inverse operation satisfies $E'E = EE' = I$ by Lemma 4.2.10. *(For (T2) one uses λ^{-1}: the field once more.)* ■

How it is computed: solving $AX = I$

The inverse is not guessed: it is **found** by solving $AX = I$, which is n systems with the same matrix A (one per column of I), solved simultaneously by elimination (**Gauss–Jordan**: reduce $[A \mid I]$ to $[I \mid A^{-1}]$, if possible).

The target here is **not the echelon form** of Session 1, but the identity, which is considerably more. That it can be reached needs a justification — and the justification also says exactly when it fails.

Proposition 4.2.15b (why Gauss–Jordan reaches I). Let A be square $n \times n$ and let A' be an echelon form of it. If A' has n pivots, then elementary operations take A' all the way to I_n. If it has fewer than n pivots, the identity cannot be reached — and A is not invertible.

Proof. Suppose there are n pivots. The pivot of row i lies strictly to the right of the pivot of row $i-1$, so by induction it falls in a column $\geq i$; since n pivots are spread over n columns, it falls exactly in column i: **the pivots occupy the diagonal**, and A' is upper triangular with nonzero diagonal. Then:

1. with (T2), divide each row i by its pivot p_i —**the field again**, $p_i \neq 0$— and the diagonal becomes all ones;
2. with (T3), **from the bottom up**: the pivot of row n kills every entry above it in column n; then the pivot of row $n-1$ does the same in its column, and so on up to row 1. Subtracting a multiple of row k only

modifies columns $\geq k$ (to the left of pivot k that row is zero), so it never reintroduces anything into the columns already cleared.

At the end each column has a 1 on the diagonal and zeros elsewhere: the result is I_n.

If there are fewer than n pivots, some row of A' is zero, so $A'x = 0$ has a nontrivial solution and so does $Ax = 0$ (Theorem 4.1.8). Were there a B with $BA = I$, that $x \neq 0$ would satisfy $x = BAx = B0 = 0$, absurd: A is not invertible. ■

> Note how the work splits: step 1 is what turns the echelon form into a normalized one, and step 2 is what cleans it *above* — together, the reduced echelon form (Def. 4.1.12), which for a square matrix with n pivots **is** the identity.

Example 4.2.16. $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$:

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{F_1 \leftrightarrow F_2} \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow{F_2 - 2F_1} \left[\begin{array}{cc|cc} 1 & 1 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array}\right] \xrightarrow[F_1 - F_2]{F_2 \cdot (-1)} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

Hence $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$. *Check:* $AA^{-1} = I$ ✓.

Remark 4.2.17. Read together with Corollary 4.2.11: if Gauss–Jordan takes A to I, then $E_k \cdots E_1 A = I$, that is, the chain of elementary matrices is a **left inverse of** A. By the result of Session 3 ("one side suffices", Corollary 4.3.24), that chain coincides with A^{-1}: $A^{-1} = E_k \cdots E_1$. That *every* invertible matrix is a product of elementary matrices will also be formalized there.

Remark 4.2.18 (what it is for). If A is invertible, $Ax = b \iff x = A^{-1}b$. For *computing* by hand, direct Gaussian elimination on $[A|b]$ is usually preferable; the inverse is above all a **conceptual** tool.

Exercises

1. ★ Write in the form $Ax = b$ the system $\{2x - y + z = 1;\ x + 3z = 0\}$, identify A, x, b and their sizes, and check the reading by columns (Proposition 4.2.3) with the solution of your choice.
2. ★ With $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$, compute AB and BA and check that they do not coincide. Find **your own** pair of matrices with $AB \neq BA$.
3. ★ Compute by Gauss–Jordan the inverse of $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ and check $AA^{-1} = I$.

4. ★ Let A be a matrix with two equal columns. Prove that $Ax = 0$ has some nontrivial solution and deduce, using the Remark following 2.13 (or directly: if there were a B with $BA = I$, then $x = BAx = 0$), that A is not invertible.
5. Prove the distributive property $B(A + A') = BA + BA'$ with explicit sums.
6. Prove $(BA)^\top = A^\top B^\top$. *(Hint: $\left((BA)^\top\right)_{ij} = (BA)_{ji} = \sum_k b_{jk} a_{ki}$; rewrite it as row i of $A^\top$ times column j of $B^\top$.)*
7. Complete Lemma 4.2.10 for types (T1) and (T2).
8. Prove that the identity is the only $n \times n$ matrix M such that $Mx = x$ for every x. *(Hint: Lemma 4.2.6.)*
9. Let A be invertible. Prove that $A^\top$ is invertible with $(A^\top)^{-1} = (A^{-1})^\top$. *(Hint: transpose $AA^{-1} = I$.)*

Going further and references

Going further (powers and iterated processes). If a process is modelled as $x \mapsto Ax$, iterating it k times is $x \mapsto A^k x$. This idea —powers of matrices— governs Markov chains, linear recurrences (Fibonacci included) and Google's original PageRank. See Strang, *Introduction to Linear Algebra*, where it is a recurring theme.

References. Product and composition: Hefferon, *Linear Algebra*, Three.IV; Strang, ch. 2. Elementary matrices and the inverse by Gauss–Jordan: Strang, §2.5; Hefferon, Three.IV–V. Transpose: Strang, §2.7.

Lecture Notes · Linear Algebra · Session 3 — Vector spaces, independence and rank

The most structural session of the module, with the major proofs: the exchange lemma (dimension is well defined), the equality of row rank and column rank, and the characterization of invertibility by rank — including the promise left pending from Session 2 (checking the inverse on one side suffices). Exercises and a Going further section at the end.

3.1 Vector spaces

Definition 4.3.1 (vector space). A **vector space over a field** K is a set V with two operations,

$$+ : V \times V \to V \quad \text{(vector addition)}, \qquad \cdot : K \times V \to V \quad \text{(scalar multiplication)},$$

satisfying, for all $u, v, w \in V$ and $a, b \in K$: 1. *(Addition)* associative and commutative; there is a **zero vector** $0 \in V$ with $v + 0 = v$; every v has an

opposite $-v$ with $v+(-v)=0$. 2. *(Scalars)* $a(u+v)=au+av$; $(a+b)v=av+bv$; $(ab)v=a(bv)$; $1\cdot v=v$.

We call the elements of V **vectors**, whatever they may be: the name denotes the role they play, not their appearance.

> **Example 4.3.2 (vectors wear many disguises).**
>
> 1. K^n, the n-tuples, with componentwise operations. It will be our working model.
> 2. $K[x]_{\leq n}$, the **polynomials of degree** $\leq n$ with coefficients in K, with the usual sum and multiplication by a constant.
> 3. The $m\times n$ **matrices** over K, with entrywise sum and scalar multiplication.
> 4. The **functions** $f:\mathbb{R}\to\mathbb{R}$, with $(f+g)(t)=f(t)+g(t)$ and $(af)(t)=a\,f(t)$.
> 5. The **sequences** $(x_0,x_1,x_2,\dots)$ satisfying a fixed linear recurrence, e.g. $x_{k+2}=x_{k+1}+x_k$ (the "Fibonacci-type sequences" form a vector space!).
>
> Verifying the axioms in each case is routine but instructive (exercise 1).

Proposition 4.3.3 (consequences of the axioms). In every vector space: (i) $0\cdot v=0$; (ii) $a\cdot 0=0$; (iii) $(-1)v=-v$; (iv) if $av=0$ then $a=0$ or $v=0$.

Proof. (i) $0v=(0+0)v=0v+0v$; adding $-(0v)$ to both sides, $0=0v$. (iv) If $a\neq 0$, multiplying $av=0$ by a^{-1} (the field!): $v=a^{-1}(av)=a^{-1}0=0$ using (ii). The proofs of (ii) and (iii) are exercise 2. ■

3.2 Subspaces

Definition 4.3.4. A nonempty subset $S\subseteq V$ is a **subspace** if it is **closed** under the two operations: $u,v\in S\Rightarrow u+v\in S$, and $a\in K, v\in S\Rightarrow av\in S$.

A subspace is, with the inherited operations, a vector space in its own right (it contains 0: take $v\in S$ and $0=0\cdot v\in S$).

> **Example 4.3.5.** In $\mathbb{R}^2$: every line **through the origin** is a subspace. A line that does *not* pass through the origin is not (it does not contain 0); nor is the unit circle (it is not closed under addition).

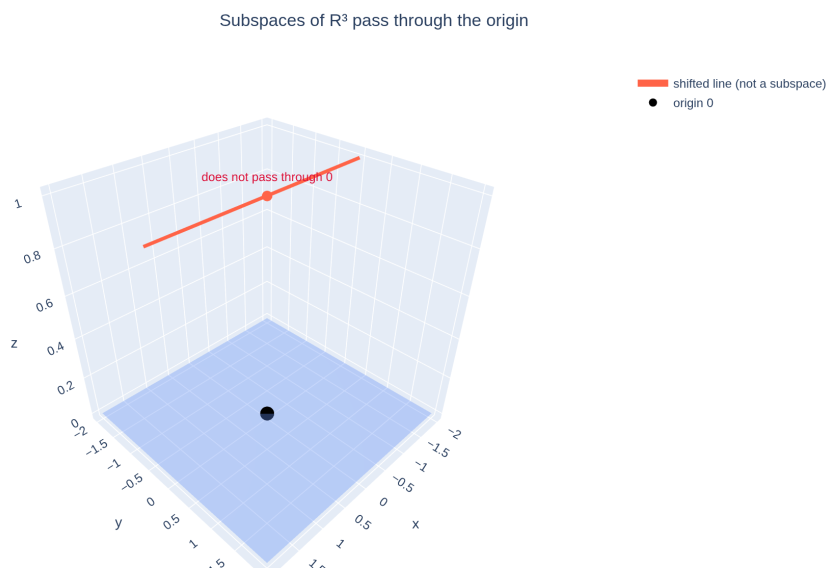

Figure 4: A subspace of $\mathbb{R}^3$ passes through the origin (a plane, here); a shifted line that misses 0 is not a subspace.

[see interactive version]

Proposition 4.3.6 (the solutions of the homogeneous system form a subspace). For any $m \times n$ matrix A, the set $\{x \in K^n : Ax = 0\}$ is a subspace of K^n.

Proof. It is nonempty (0 is a solution). If $Au = 0$ and $Av = 0$, by Proposition 4.2.2: $A(u + v) = Au + Av = 0$ and $A(\lambda u) = \lambda(Au) = 0$. ■

> The structure is not decorative: it appears on its own in the objects we were already handling.

3.3 Linear combinations, span and generating sets

Definition 4.3.7. A **linear combination** of $v_1, \dots, v_k \in V$ is any vector of the form $a_1 v_1 + \dots + a_k v_k$ with $a_i \in K$. The set of **all** linear combinations of $v_1, \dots, v_k$ is called their **span** (or *subspace generated by them*), $\mathrm{span}(v_1, \dots, v_k)$. We say that $v_1, \dots, v_k$ **generate** V if $\mathrm{span}(v_1, \dots, v_k) = V$; such a set is a **generating set** of V.

Proposition 4.3.8. $\mathrm{span}(v_1, \dots, v_k)$ is always a subspace of V (and it is the smallest one containing the v_i).

Proof. The sum of two linear combinations is a linear combination (grouping coefficients: $\sum a_i v_i + \sum b_i v_i = \sum (a_i + b_i) v_i$), and a multiple of a combination is one too ($\lambda \sum a_i v_i = \sum (\lambda a_i) v_i$). It is the smallest: any subspace containing the v_i must contain, by closure, all their combinations. ■

3.4 Linear dependence and independence

Definition 4.3.9. The vectors $v_1, \ldots, v_k$ are **linearly dependent** if there is a **nontrivial** linear combination equal to the zero vector:

$$a_1 v_1 + \cdots + a_k v_k = 0 \quad \text{with some } a_i \neq 0.$$

They are **linearly independent** if the only combination giving 0 is the trivial one ($a_1 = \cdots = a_k = 0$).

Proposition 4.3.10 (the equivalent form). $v_1, \ldots, v_k$ are dependent **if and only if** one of them is a linear combination of the others.

Proof. ($\Leftarrow$) If $v_i = \sum_{j \neq i} b_j v_j$, then $\sum_{j \neq i} b_j v_j + (-1)\, v_i = 0$ is a nontrivial combination (the coefficient of v_i is $-1 \neq 0$). ($\Rightarrow$) If $\sum_j a_j v_j = 0$ with $a_i \neq 0$, we solve for v_i **dividing by** a_i (the field!): $v_i = \sum_{j \neq i} (-a_i^{-1} a_j)\, v_j$. ■

Remark 4.3.10b (geometric reading: the dependent one widens nothing). If $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$, then

$$\operatorname{span}(v_1, \ldots, v_k) = \operatorname{span}(v_1, \ldots, v_{k-1}).$$

Indeed: $\operatorname{span}(v_1, \ldots, v_{k-1})$ is a subspace (Proposition 4.3.8) containing the k vectors, hence it contains **all** their linear combinations, that is, the span of the k of them; the other inclusion is obvious. Read backwards: vectors are independent exactly when **each one contributes a direction the others cannot reach**, and as soon as one stops contributing it the span stalls.

Method 4.3.11 (deciding independence = solving a homogeneous system). In K^m, the coefficients of the combination become the **unknowns** of a system; as such we denote them $x_1, \ldots, x_k$. Setting up $x_1 v_1 + \cdots + x_k v_k = 0$ is, by the reading by columns (Proposition 4.2.3), setting up $Ax = 0$ with $x = (x_1, \ldots, x_k) \in K^k$ and A the matrix whose **columns** are the v_i. Therefore: *independent* $\iff$ *the homogeneous system* $Ax = 0$ *has only the trivial solution*, and that is decided by Gaussian elimination. *(The change of letter is not cosmetic: it spares having to read the resulting echelon form while translating the names of the unknowns.)*

> **Example 4.3.12.** Are $(1,1,0)$, $(1,0,1)$, $(0,1,1)$ independent in $\mathbb{R}^3$? We reduce $\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$: three pivots, only the trivial solution — **independent**. (Over $\mathbb{Z}/2\mathbb{Z}$ the answer changes: the sum of the three is $(0,0,0)$, dependent! The field matters.)

3.5 Bases and dimension

Definition 4.3.13. A **basis** of V is a set of vectors that is **linearly independent** and **generating**.

> **Example 4.3.14.** The **standard basis** $e_1, \dots, e_n$ of K^n. In $K[x]_{\leq n}$: $\{1, x, x^2, \dots, x^n\}$ ($n+1$ vectors!). In the $m \times n$ matrices: the mn matrices with a single 1.

Lemma 4.3.15 (exchange lemma, Steinitz). If $\{w_1, \dots, w_m\}$ generates V and $\{v_1, \dots, v_k\}$ is linearly independent in V, then $k \leq m$.

Proof. We will see that the v_i can **replace** the w_j one by one without losing the generating property. Since $\{w_j\}$ generates, $v_1 = \sum_j a_j w_j$ with some $a_j \neq 0$ ($v_1 \neq 0$ by independence); reordering, $a_1 \neq 0$. Solving for w_1 (dividing by a_1 — the field), $w_1 \in \operatorname{span}(v_1, w_2, \dots, w_m)$, so $\{v_1, w_2, \dots, w_m\}$ generates V. We iterate: assuming $\{v_1, \dots, v_i, w_{i+1}, \dots, w_m\}$ generates, we write v_{i+1} as a combination of them; **some coefficient of some w_j ($j > i$) is nonzero** —otherwise v_{i+1} would be a combination of $v_1, \dots, v_i$, contradicting independence (Proposition 4.3.10)—, and that w_j is replaced by v_{i+1} as before. If we had $k > m$, the process would exhaust the w's at step m and leave $\{v_1, \dots, v_m\}$ generating, so that v_{m+1} would be a combination of $v_1, \dots, v_m$: contradiction. Hence $k \leq m$. ■

Theorem 4.3.16 (dimension is well defined). If V has a finite basis, all its bases have the same number of elements.

Proof. Let B and B' be bases with $|B| = m$, $|B'| = k$. Since B generates and B' is independent, $k \leq m$ (Lemma 4.3.15); exchanging roles, $m \leq k$. ■

Definition 4.3.17. That common number is the **dimension** of V, $\dim V$. Thus, $\dim K^n = n$, $\dim K[x]_{\leq n} = n+1$, $\dim(m \times n \text{ matrices}) = mn$.

Proposition 4.3.18 (coordinates). If $\{b_1, \dots, b_n\}$ is a basis of V, every $v \in V$ can be written in a **unique** way as $v = c_1 b_1 + \cdots + c_n b_n$. The scalars $(c_1, \dots, c_n) \in K^n$ are the **coordinates** of v in that basis.

Proof. Existence: the basis generates. Uniqueness: if $\sum c_i b_i = \sum c_i' b_i$, subtracting, $\sum (c_i - c_i') b_i = 0$; by independence all the $c_i - c_i' = 0$. ■

> **In finite dimension, everything reduces to K^n.** Once a basis is fixed, the assignment $v \mapsto (c_1, \dots, c_n)$ is a one-to-one correspondence $V \leftrightarrow K^n$ that **respects the operations** (the coordinates of $u + v$ are the sum of the coordinates, etc. — exercise 6). Working in K^n loses no generality. The technical name of this correspondence will arrive in Session 5.

Lemma 4.3.18b (existence and extension of bases). Let V have finite dimension n. 1. Every linearly independent family in V can be **extended** to a basis of V. 2. Every **subspace** $S \subseteq V$ has a (finite) basis, and $\dim S \leq n$.

Proof. (1) If the independent family does not generate V, there is a v outside its span; upon adding it the family **remains independent** (in a null combination, the coefficient of v must be 0 —otherwise v could be solved for as a combination of the others, Proposition 4.3.10, against $v \notin$ span— and then the remaining coefficients are 0 by the previous independence). The process must terminate: no independent family in V exceeds n elements (Lemma 4.3.15, against a basis of V). At termination, the family is independent and generating: a basis. (2) The same process **inside** S, starting from the empty family (or from any nonzero $v \in S$): as long as what has been built does not generate S, add a vector of S outside the span. It terminates, by the same bound n, in a basis of S with at most n elements. ■

> This lemma closes two gaps that we will use without further mention: that it makes sense to speak of the dimension of a subspace (e.g. in the definition of rank, §3.6) and that "taking a basis of the kernel and extending it" is legitimate (rank–nullity theorem, Session 5).

3.6 Rank

Let A be an $m \times n$ matrix. Its rows live in K^n and its columns in K^m; each family generates its span.

Definition 4.3.19. The **row rank** of A is dim(span of the rows); the **column rank**, dim(span of the columns).

A priori these are **two different numbers**: they measure things in different spaces and there is no evident reason for them to coincide. That they do coincide is a **theorem**.

Lemma 4.3.20. Elementary row operations **do not change the span of the rows**.

Proof. Each new row is a linear combination of the old ones (evident for the three types), so the new span is contained in the old one; since each operation is reversible (Theorem 4.1.8), also conversely. ■

Lemma 4.3.21. Elementary row operations **do not change the dependence relations among the columns**: $Ax = 0 \iff A'x = 0$ when A' results from A by row operations.

Proof. A dependence $\sum x_j c_j = 0$ among the columns is exactly a solution of the homogeneous system $Ax = 0$ (Proposition 4.2.3), and row operations preserve the solution set (Theorem 4.1.8). ■

Proposition 4.3.22 (in an echelon matrix). Let R be in echelon form with r pivots. Then: (i) its r nonzero rows are independent, so the row rank of R is r; (ii) its r pivot columns are independent and the remaining ones are combinations of them, so the column rank of R is r.

Proof. (i) In a null combination of the nonzero rows, looking at the column of the first pivot: only the first row has a nonzero entry there, so its coefficient is 0; one keeps iterating with each pivot. (ii) The pivot columns follow the "staircase" pattern (each one opens a new nonzero position below the previous ones): the same argument gives independence. A column without a pivot: the system $Rx = 0$ allows solving (back substitution) for each pivot variable in terms of the free ones; setting the free variable of that column to 1 and the other free ones to 0 yields a dependence expressing it as a combination of the pivot columns. ■

Theorem 4.3.23 (row rank = column rank). For every matrix A, the two ranks coincide — the common number is called the **rank** of A, $\operatorname{rank} A$, and it is the **number of pivots** of any echelon form of A.

Proof. Reduce A to an echelon form R with r pivots. Row rank: the row span does not change (Lemma 4.3.20) and that of R has dimension r (Proposition 4.3.22.i). Column rank: the dependence relations among columns do not change (Lemma 4.3.21), so a set of columns of A is independent exactly when the corresponding columns of R are; by 4.3.22.ii the maximal independent family has r elements in both cases. Hence both ranks equal r. ■

Corollary 4.3.24 (invertibility and rank; the promise of Session 2). For a square $n \times n$ matrix A, the following are equivalent: 1. $\operatorname{rank} A = n$ (full rank); 2. the reduced echelon form of A is I; 3. A is a product of elementary matrices; 4. A is invertible; 5. $Ax = 0$ has only the trivial solution (independent columns).

Moreover, **one side suffices**: if $BA = I$ (or $AB = I$), then A is invertible and $B = A^{-1}$.

Proof. (1⇒2) With n pivots in n columns, the reduced form has a 1 in each diagonal position and zeros elsewhere: it is I. (2⇒3) $E_k \cdots E_1 A = I$ with the E_i elementary; solving with inverses of elementary matrices (Proposition 4.2.15.2), $A = E_1^{-1} \cdots E_k^{-1}$, a product of elementary matrices. (3⇒4) A product of invertible matrices is invertible (2.15.1). (4⇒5) If $Ax = 0$, then $x = A^{-1}Ax = 0$. (5⇒1) Independent columns ⇒ column rank $= n$.

One side suffices: suppose $BA = I$. If $Ax = 0$, then $x = (BA)x = B(Ax) = 0$: (5) holds, so A is invertible. Then $B = B(AA^{-1}) = (BA)A^{-1} = A^{-1}$, and in particular $AB = I$. The case $AB = I$ is obtained by applying the above to B. ■

> The rank measures *how much independent information* the matrix contains. The systematic application to systems —when a solution exists and how many there are— is Session 4.

Exercises

1. ★ Verify the vector space axioms in two of the items of Example 4.3.2 (recommended: polynomials and Fibonacci sequences).
2. Prove (ii) and (iii) of Proposition 4.3.3.
3. ★ Decide whether the following are subspaces of the indicated spaces, proving it: (a) $\{(x, y) : x + y = 0\} \subseteq \mathbb{R}^2$; (b) $\{(x, y) : x + y = 1\} \subseteq \mathbb{R}^2$; (c) the polynomials in $K[x]_{\leq 3}$ with $p(1) = 0$; (d) the 2×2 matrices with

determinant… we do not have determinants yet: with **trace** (sum of the diagonal) zero.

4. ★ Decide by Method 4.3.11 whether $(1,2,3),(4,5,6),(7,8,9)$ are independent in $\mathbb{R}^3$. Find a basis and the dimension of their span.
5. ★ Compute the rank of $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 \\ 3 & 6 & 1 & 2 \end{pmatrix}$ and give explicitly a dependence among its columns.
6. Check that coordinates respect the operations: if $v \leftrightarrow (c_i)$ and $w \leftrightarrow (d_i)$ in a fixed basis, then $v + w \leftrightarrow (c_i + d_i)$ and $av \leftrightarrow (ac_i)$.
7. **(Diverse spaces.)** Find a basis and the dimension of: (a) $\{p \in K[x]_{\le 2} : p(0) = 0\}$; (b) the **symmetric** 2×2 matrices ($A^{\mathsf{T}} = A$); (c) the space of solutions of the recurrence $x_{k+2} = x_{k+1} + x_k$ *(hint: a sequence is determined by (x_0, x_1))*.
8. **(Structural counterfactual.)** Can 3 vectors in K^2 be independent? Prove that they cannot, citing the exact result that forbids it.
9. Give a 3×3 matrix of rank 2 and check, via Corollary 4.3.24, that it is not invertible (exhibit a nontrivial solution of $Ax = 0$).
10. True or false? "If the rows of a square matrix are independent, so are its columns." Justify with the relevant result.

Going further and references

Going further (infinite dimension). The space of *all* polynomials $K[x]$, or that of all functions $\mathbb{R} \to \mathbb{R}$, has no finite basis: they are **infinite-dimensional**. The theory extends (every basis still has the same cardinality), but it requires tools from set theory. See Axler, *Linear Algebra Done Right*, ch. 2 (notes), or Aluffi, ch. VI.

References. Spaces and subspaces: Axler, ch. 1; Hefferon, Two.I. Independence, bases, dimension (and the exchange lemma): Axler, ch. 2; Hefferon, Two.II–III. Row rank = column rank: Hefferon, Two.III.3; Strang, §3.3 (with the "four subspaces" reading).

Lecture Notes · Linear Algebra · Session 4 — Rouché–Capelli and parametrization

Rank applied to systems: the complete consistency criterion (with proof), the parametrization of the solutions —where *the proof is the method*—, the structure "general = particular + kernel", and the one-sided inverses of non-square matrices. Exercises and a Going further section at the end.

4.1 Rank decides consistency

In all that follows, A is $m \times n$ over K, $b \in K^m$, and n is the **number of unknowns** (= columns of A). We denote by $[A \mid b]$ the augmented matrix.

Proposition 4.4.1. The system $Ax = b$ is consistent **if and only if** $b \in$ span(columns of A).

Proof. By the reading by columns (Proposition 4.2.3), $Ax = x_1 c_1 + \cdots + x_n c_n$. That a solution exists is exactly that b is a linear combination of the columns. ■

Lemma 4.4.2. $b \in \operatorname{span}(\text{columns of } A) \iff \operatorname{rank}[A \mid b] = \operatorname{rank} A$.

Proof. Let $S = \operatorname{span}(c_1, \dots, c_n)$ and $S' = \operatorname{span}(c_1, \dots, c_n, b)$, the column spans of A and of $[A|b]$; always $S \subseteq S'$. ($\Rightarrow$) If $b \in S$, every combination of columns and b can be rewritten as a combination of columns: $S' = S$, and the (column) ranks coincide. ($\Leftarrow$) If $b \notin S$, we claim that $\dim S' > \dim S$: taking a basis $w_1, \dots, w_r$ of S, the family $w_1, \dots, w_r, b$ is independent —in a null combination $\sum a_i w_i + a\, b = 0$, if $a \neq 0$ we could solve for b (dividing by a: the field) as a combination of the w_i, against $b \notin S$; hence $a = 0$ and then all the $a_i = 0$ by independence—. Thus $\operatorname{rank}[A|b] \geq r + 1 > \operatorname{rank} A$. ■

Lemma 4.4.2b (uniqueness in the homogeneous case, rectangular version). For any $m \times n$ matrix A, the following are equivalent: (i) $Ax = 0$ has only the trivial solution; (ii) the n columns of A are linearly independent; (iii) $\operatorname{rank} A = n$.

Proof. (i) $\iff$ (ii): a nontrivial solution of $Ax = 0$ is exactly a nontrivial dependence among the columns (reading by columns, Proposition 4.2.3, as in Method 4.3.11). (ii) $\iff$ (iii): the columns are n vectors; they are independent exactly when the dimension of their span —the column rank— is n. ■

Theorem 4.4.3 (Rouché–Capelli, known in Spanish textbooks as the Rouché–Frobenius theorem). For the system $Ax = b$ with n unknowns: 1. It is **consistent** $\iff$ $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. 2. If it is consistent: it has a **unique solution** $\iff$ $\operatorname{rank} A = n$, and **infinitely many** $\iff$ $\operatorname{rank} A < n$, with $n - \operatorname{rank} A$ degrees of freedom (§4.2).

Proof. (1) It is the combination of 4.4.1 and 4.4.2. (2) Suppose $Ax = b$ consistent and let x_0 be a solution. If u, v are solutions, $A(u - v) = Au - Av = b - b = 0$: the difference solves the homogeneous system. Conversely, if $Ah = 0$, then $A(x_0 + h) = b$. Therefore *the solution is unique exactly when the homogeneous system has only the trivial solution*, which by **Lemma 4.4.2b** is equivalent to $\operatorname{rank} A = n$. The count of degrees of freedom is proved in Theorem 4.4.6. ■

> This theorem **rigorously justifies** the intuitive reading of Session 1: a row $0 = c$ ($c \neq 0$) betrays $\operatorname{rank}[A|b] > \operatorname{rank} A$; and uniqueness depends on whether every column has a pivot — not on the rows $0 = 0$ (Remark of Session 1).

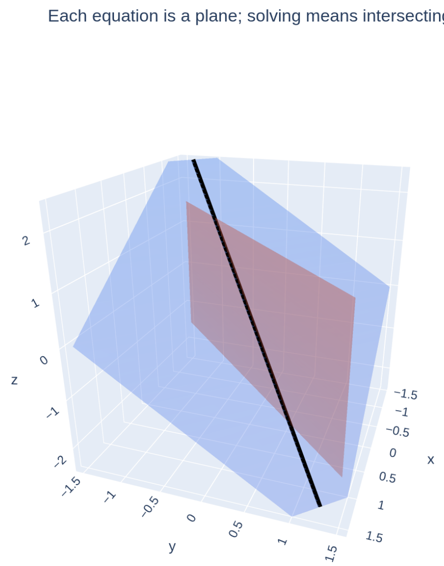

Figure 5: Each equation of a system is a plane in $\mathbb{R}^3$; the solution is their intersection (here, a line).

[see interactive version]

4.2 Parametrization: the proof is the method

Reduce $[A \mid b]$ to echelon form (consistent, $r = \operatorname{rank} A$ pivots). The unknowns split into two classes: the **pivot** unknowns (their column has a pivot) and the **free** ones (the others, $n - r$).

Method 4.4.4 (parametrization). Assign to each free unknown a **parameter** ($x_j = t_j$) and solve for the pivot unknowns, from the bottom up, in terms of the parameters (dividing by the pivots: the field). The result is the **general solution**: an expression

$$x = x_p + t_1 h_1 + t_2 h_2 + \cdots + t_{n-r}\, h_{n-r}, \qquad t_i \in K,$$

where x_p is the solution obtained with all parameters set to zero and the h_i are fixed vectors (the "coefficients" of each parameter).

Theorem 4.4.5 (the method is exhaustive and without repetitions). The assignment $(t_1, \dots, t_{n-r}) \mapsto x$ of Method 4.4.4 is a **bijection** between K^{n-r} and the solution set.

Proof. Every assignment gives a solution: by construction, the solved-for values satisfy each echelon equation, and the echelon system is equivalent to the original one (Theorem 4.1.8). *Every solution is reached:* given a solution s, its values in the free positions determine some parameters; the pivot unknowns of s are forced by back substitution (at each step, the solved-for value is unique: one divides by a nonzero pivot), so s coincides with the solution the method assigns to those parameters. *No repetitions:* two distinct tuples of parameters differ in some free position, and the corresponding solutions differ in that coordinate. ■

> Hence the slogan: the relation "pivots $\leftrightarrow$ determined, no pivot $\leftrightarrow$ free, $n-r$ degrees of freedom" **needs no separate proof: the proof is the very method** of reducing to echelon form and solving.

Corollary 4.4.5b (exactly how many solutions there are). Theorem 4.4.5 does not merely say that the solutions can be parametrized: being a **bijection**, it says there are as many solutions as tuples of parameters. Hence, for a consistent system with $r = \mathrm{rg} A$ and n unknowns,

$$\#\{\text{solutions}\} \;=\; |K^{n-r}| \;=\; |K|^{n-r}.$$

In particular:

- if K is **infinite** and $r < n$, the solutions are **infinitely many** — that is the exact sense of the "infinitely many" we have been saying since Session 1, now quantified;
- if K is **finite** with q elements, there are **exactly** q^{n-r}: over $\mathbb{Z}/7\mathbb{Z}$ with one free parameter there are seven solutions, not one more; over $\mathbb{Z}/2\mathbb{Z}$ with three parameters, eight;
- and $r = n$ gives $|K|^0 = 1$ **over any field**: uniqueness does not depend on the size of K.

> **Why it is worth saying this way.** "Consistent indeterminate" means *more than one solution* (Definition 4.1.6), not *infinitely many*: the latter follows only if K is infinite. Since the course convention is $K = \mathbb{R}$ unless stated otherwise, saying "infinitely many" is correct in nearly everything we do — but the statement to retain is $|K|^{n-r}$, which covers both cases and **need not be unlearned** when working over $\mathbb{Z}/p\mathbb{Z}$, where the course has been since Module 1.

Definition and Theorem 4.4.6 (the kernel and its dimension). The solution set of $Ax = 0$ —a subspace of K^n by Proposition 4.3.6— is called the **kernel** of A. Its dimension is

$$\dim(\text{kernel of } A) = n - \mathrm{rank} A.$$

Proof. We apply the method to the homogeneous system ($b = 0$; then $x_p = 0$): the general solution is $t_1 h_1 + \cdots + t_{n-r} h_{n-r}$, so that $h_1, \ldots, h_{n-r}$ **generate** the kernel. They are moreover **independent**: the vector h_i has a 1 in the position of the i-th free unknown and 0 in the other free positions (that is how it was built), so in a null combination $\sum t_i h_i = 0$, looking at each free position, all the $t_i = 0$. Hence they form a **basis** of the kernel, which has dimension $n - r$. ■

4.3 Structure of the solution set

Theorem 4.4.7 (general = particular + kernel). If x_p is any solution of $Ax = b$, then

$$\{\text{solutions of } Ax = b\} = x_p + \{\text{kernel of } A\} := \{\, x_p + h : Ah = 0 \,\}.$$

Proof. ($\supseteq$) $A(x_p + h) = Ax_p + Ah = b + 0 = b$. ($\subseteq$) If $As = b$, then $A(s - x_p) = b - b = 0$, so $h := s - x_p$ lies in the kernel and $s = x_p + h$. ∎

Geometric reading. The solution set is **the kernel translated** by x_p. Mind the language: the subspace (the kernel) *does not move*; what happens is that the solutions **sweep out** that translate as the parameters vary.

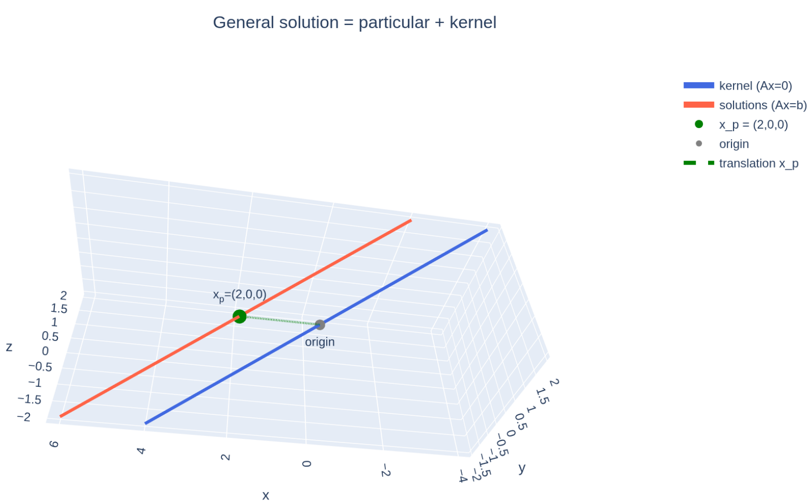

Figure 6: The structure of the solutions: the kernel (solutions of $Ax = 0$, through the origin) translated by a particular solution x_p gives all the solutions of $Ax = b$ — an affine subspace.

[see interactive version]

> **Example 4.4.8.** $\begin{cases} x_1 + x_2 + x_3 = 2 \\ x_2 - x_3 = 0 \end{cases}$ (already in echelon form, $r = 2$, $n = 3$): free $x_3 = t$; solving, $x_2 = t$, $x_1 = 2 - 2t$:
>
> $$(x_1, x_2, x_3) = (2, 0, 0) + t\,(-2, 1, 1).$$
>
> The kernel is the **line through the origin** with direction $(-2, 1, 1)$; the solutions, that line **translated** to the point $(2, 0, 0)$ — a line that no longer passes through the origin. *(This picture — "linear object + point it passes through" — will reappear as a foundational idea in the Geometry module.)*

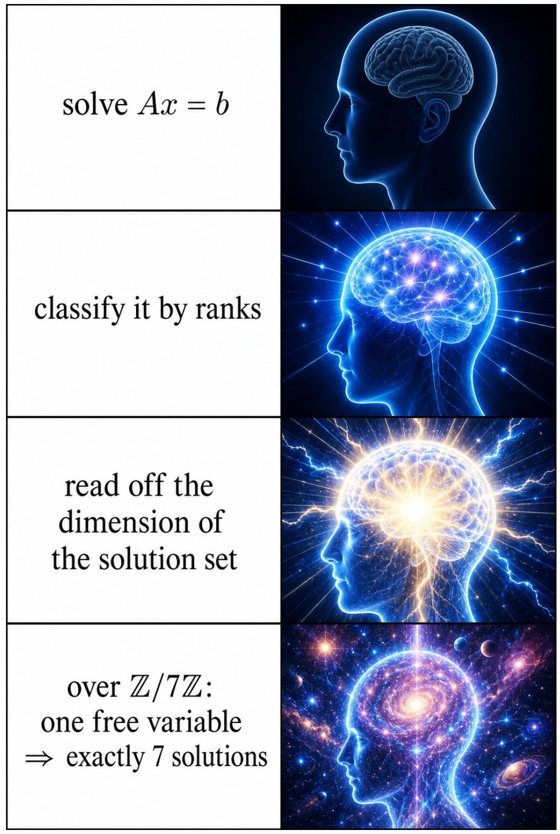

Figure 7: The session's ladder, meme edition: solving one concrete system is the first rung; classifying it by ranks (Rouché–Capelli) the second; reading off the **dimension** of the solution set ($n - r$ parameters) the third; and at the top, **counting**: each parameter ranges over K, so over $\mathbb{Z}/7\mathbb{Z}$ one free variable gives exactly 7 solutions.

4.4 One-sided inverses: the non-square case

For A of size $m \times n$ with $m \neq n$ no two-sided inverse can exist, but there may be one **on one side**, and the rank says exactly when.

Proposition 4.4.9. Let A be of size $m \times n$. 1. There exists X $(n \times m)$ with $AX = I_m$ (a **right inverse**) $\iff$ $\operatorname{rank} A = m$ (rank = number of rows). 2. There exists Y $(n \times m)$ with $YA = I_n$ (a **left inverse**) $\iff$ $\operatorname{rank} A = n$ (rank = number of columns).

Proof. (1) $AX = I_m$ amounts to solving the m systems $Ax = e_i$ ($i = 1, \dots, m$, columns of I_m). All consistent $\iff$ each $e_i \in \operatorname{span}(\text{columns of } A)$ (Proposition 4.4.1) $\iff$ that span is **all** of K^m $\iff$ (column) rank $= m$. (2) $YA = I_n$ $\iff$

$A^\mathsf{T} Y^\mathsf{T} = I_n$ (transpose, Session 2, ex. 6), which by part (1) applied to A^T (of size $n \times m$) is equivalent to $\operatorname{rank} A^\mathsf{T} = n$. And $\operatorname{rank} A^\mathsf{T} = \operatorname{rank} A$, because transposing interchanges rows and columns and **row rank and column rank coincide** (Theorem 4.3.23) — here is an unexpected dividend of that theorem. ■

Corollary 4.4.10. Since $\operatorname{rank} A \le \min(m, n)$: if $m \neq n$, **at most one side** is possible ($m < n$: only right; $m > n$: only left). And if $m = n$ and one side exists, both exist and coincide (Corollary 4.3.24).

Exercises

1. ★ Discuss according to $a \in \mathbb{R}$, using Rouché–Capelli (comparing ranks, without solving completely):

$$\begin{cases} x + y + z = 1 \\ x + 2y + z = 2 \\ 2x + 3y + (a^2 - 2)\, z = a + 1 \end{cases}$$

2. ★ Parametrize all the solutions of $\begin{cases} x_1 + 2x_2 - x_3 + x_4 = 3 \\ x_2 + x_3 - x_4 = 1 \end{cases}$, identify the free unknowns, write the solution as $x_p + t_1 h_1 + t_2 h_2$ and check that h_1, h_2 solve the homogeneous system.
3. ★ In the previous exercise, give a **basis of the kernel** and verify $\dim = n - r$.
4. ★ Draw the solution set of $\{x + y = 2\}$ in $\mathbb{R}^2$: the kernel of $(1\ 1)$ and its translate. Mark x_p and the direction.
5. Reread the cases "row $0 = 0$" and "row $0 = c$" of Session 1 and translate them into statements about $\operatorname{rank} A$ and $\operatorname{rank}[A|b]$.
6. Give a 2×3 matrix with a right inverse and check that it cannot have a left inverse (what rank would it need?). Compute an explicit right inverse. Is it unique? *(Hint: count degrees of freedom in $AX = I$.)*
7. True or false? "If $Ax = b$ has a unique solution, then A is square." Justify or refute with an example.
8. ★ **(The field decides how many.)** Consider $\begin{cases} x_1 + x_2 + x_3 = 2 \\ \qquad\quad x_2 - x_3 = 0 \end{cases}$
   (a) Over $\mathbb{R}$: parametrize and check that the general solution is $(2, 0, 0) + t\,(-2, 1, 1)$, a line.
   (b) Over $\mathbb{Z}/5\mathbb{Z}$: the very same computation holds word for word, but now t ranges over only five values. **List the five solutions** and verify each one in both equations.
   (c) Check that the count agrees with Corollary 4.4.5b, $|K|^{n-r}$, and say —without computing— how many solutions it would have over $\mathbb{Z}/7\mathbb{Z}$.

Going further and references

> **Going further (what if there is no solution? — least squares).** When $Ax = b$ is inconsistent (frequent with real data: more equations than unknowns, plus noise), one looks for the x minimizing the error $\|Ax-b\|$: the method of **least squares**, the engine of linear regression in data science. It requires the dot product (Geometry module). See Strang, *Introduction to Linear Algebra*, §4.3.
>
> **References.** Rouché–Capelli and the discussion of systems: de Burgos, *Álgebra lineal y geometría cartesiana*; Hefferon, One.III. Parametrization and solutions of the homogeneous system: Hefferon, One.I.3; Strang, §3.2–3.4.

Lecture Notes · Linear Algebra · Session 5 — Linear maps

> The morphisms of vector spaces: definition and proved examples, the matrix of a map (with the determination-by-a-basis lemma and the composition proposition), kernel and image, the rank–nullity theorem with its general proof, and isomorphisms — including the lemma that the inverse of a bijective linear map is linear. Exercises and a Going further section at the end.

5.1 Definition and examples

Definition 4.5.1. Let V, W be vector spaces over the **same** field K. A map $f : V \to W$ is **linear** if it respects the two operations:

$$f(u+v) = f(u) + f(v) \quad \text{and} \quad f(av) = a\,f(v) \qquad (u, v \in V,\ a \in K),$$

or, equivalently, if it respects linear combinations: $f(au + bv) = af(u) + bf(v)$.

> **Structural lens:** linear maps are the **morphisms** of vector spaces — the "maps that respect the structure", as we anticipated with the germinal idea of Module 0.

Proposition 4.5.2. If f is linear, $f(0_V) = 0_W$.

Proof. $f(0_V) = f(0 \cdot 0_V) = 0 \cdot f(0_V) = 0_W$, using Proposition 4.3.3.i first in V and then in W. **Mind the three zeros**, written alike and not the same thing: the first 0 in $0 \cdot 0_V$ is the **scalar** zero of K; 0_V is the **zero vector** of V; and 0_W, the zero vector of W. In what follows we shall write plain 0 again whenever the context makes it clear. ■

Example 4.5.3 (with proof).

1. **Every matrix:** $f(x) = Ax$, from K^n to K^m, is linear — this is exactly Proposition 4.2.2.
2. **Projection:** $\pi : \mathbb{R}^2 \to \mathbb{R}^2$, $\pi(x, y) = (x, 0)$. *Proof:* $\pi((x, y) + (x', y')) = (x + x', 0) = \pi(x, y) + \pi(x', y')$, and $\pi(a(x, y)) = (ax, 0) = a\,\pi(x, y)$. ✓
3. **The derivative** $D : K[x]_{\leq n} \to K[x]_{\leq n}$, $D(p) = p'$. *Proof:* $(p+q)' = p'+q'$ and $(ap)' = a\,p'$ — the differentiation rules *are* linearity. ✓ *(Note: an important example in a space that is not K^m.)*
4. **Rotation** by an angle θ in $\mathbb{R}^2$: geometrically, rotating the sum of two arrows is adding the rotated arrows, and rotating a scaled arrow is scaling the rotated one — linear. (Its matrix: exercise 2.)

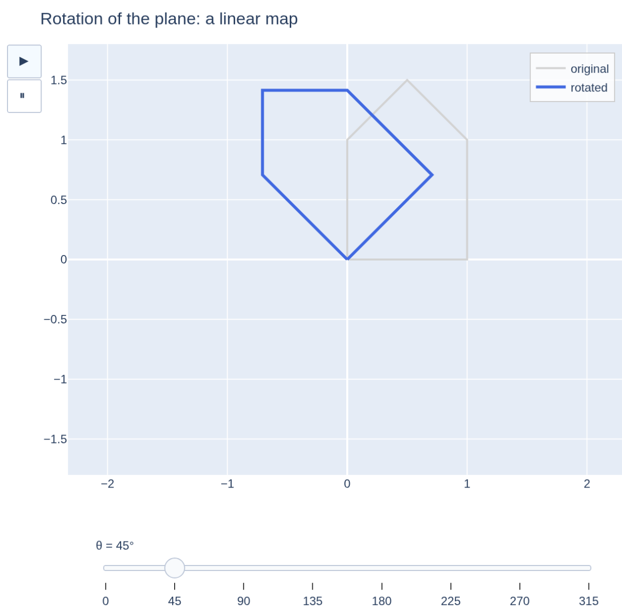

Figure 8: The rotation of the plane is a linear map: every vector is transformed by rotating it through the same angle (here, a figure and its rotated copy).

[see interactive version]

5. **Non-example:** $g(x) = x + 1$ on $\mathbb{R}$ is not linear: $g(0) = 1 \neq 0$ (Proposition 4.5.2 gives a quick test). Nor is $h(x, y) = xy$.

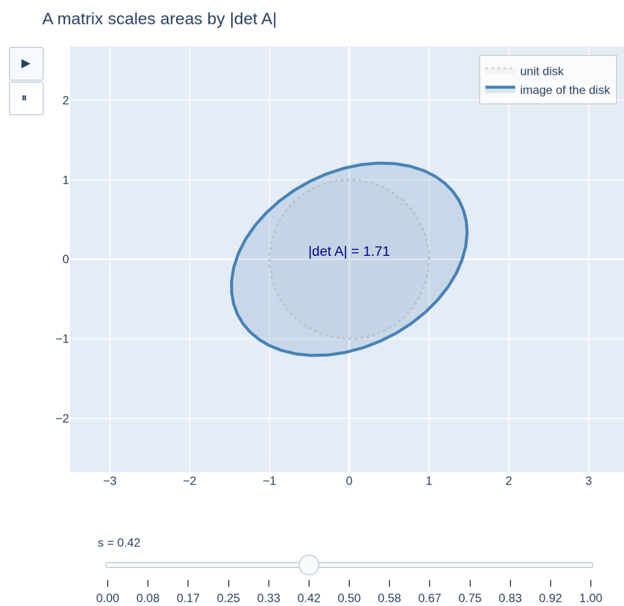

Figure 9: A 2×2 matrix transforms the unit disk into an ellipse, multiplying the area by $|\det A|$ *(the determinant* $\det A$ *is defined in Session 6; here it is enough to keep the idea that a matrix deforms the disk into an ellipse and changes the area by a fixed factor).*

[see interactive version]

5.2 The matrix of a linear map

Lemma 4.5.4 (determination by a basis). Let $\{b_1, \dots, b_n\}$ be a basis of V and $w_1, \dots, w_n \in W$ arbitrary. There exists **exactly one** linear map $f : V \to W$ with $f(b_j) = w_j$ for every j.

Proof. Uniqueness: every $v \in V$ is written uniquely as $v = \sum_j c_j b_j$ (coordinates, Proposition 4.3.18); if f is linear and $f(b_j) = w_j$, necessarily $f(v) = \sum_j c_j w_j$ — there is no choice. *Existence:* define $f(v) := \sum_j c_j w_j$ with the c_j the coordinates of v. It is linear because coordinates respect the operations (Session 3, exercise 6): the coordinates of $u + v$ are the sum of the coordinates, so $f(u + v) = f(u) + f(v)$, and likewise with scalars. ∎

Definition 4.5.5. Once bases $\{b_j\}$ of V ($\dim n$) and $\{b'_i\}$ of W ($\dim m$) are fixed, the **matrix of** f in those bases is the $m \times n$ matrix whose **column** j **holds the coordinates of** $f(b_j)$ in the basis of W. By Lemma 4.5.4, the correspondence between linear maps $V \to W$ and $m \times n$ matrices is one-to-one; in coordinates, f acts as $x \mapsto Ax$.

Proposition 4.5.6 (composition is the product). With bases fixed in

U, V, W: if $f : U \to V$ and $g : V \to W$ are linear with matrices A and B, then $g \circ f$ is linear and its matrix is BA.

Proof. The composition is linear: $(g \circ f)(au + bv) = g(af(u) + bf(v)) = a\, g(f(u)) + b\, g(f(v))$. Its matrix: in coordinates, applying f is multiplying by A and applying g is multiplying by B, so applying $g \circ f$ is $x \mapsto B(Ax) = (BA)x$ (Theorem 4.2.5); since the matrix of a map is unique (Lemma 4.2.6), the matrix of $g \circ f$ is BA. ■

> This is the formal statement promised in the video: *the why* ("row times column" is born from composing) was already seen in Session 2; here it is generalized to arbitrary maps.

5.3 Kernel and image

Definition 4.5.7. For $f : V \to W$ linear:

$$\ker f = \{v \in V : f(v) = 0\} \subseteq V \qquad \text{(kernel)},$$

$$\mathrm{im} f = \{f(v) : v \in V\} \subseteq W \qquad \text{(image)}.$$

Proposition 4.5.8. $\ker f$ is a subspace of V and $\mathrm{im} f$ is a subspace of W.

Proof. Kernel: $0 \in \ker f$ (Proposition 4.5.2); if $f(u) = f(v) = 0$, then $f(u + v) = 0 + 0 = 0$ and $f(av) = a \cdot 0 = 0$. *(The same argument as for $Ax = 0$ in Session 3 — in fact, that one was this one with $f(x) = Ax$.) Image:* nonempty ($0 = f(0)$); and it is **linearity** that closes it: $f(u) + f(v) = f(u + v) \in \mathrm{im} f$ and $a\, f(v) = f(av) \in \mathrm{im} f$. *(For a non-linear map, the image of a space need not be a subspace.)* ■

Proposition 4.5.9 (the matrix case). For $f(x) = Ax$: $\ker f = \{x : Ax = 0\}$ (the kernel of A, Definition and Theorem 4.4.6) and $\mathrm{im} f = \mathrm{span}(\text{columns of } A)$.

Proof. The first equality is the definition. The second: $Ax = x_1 c_1 + \cdots + x_n c_n$ (Proposition 4.2.3), so the values attained are exactly the linear combinations of the columns. ■

5.4 The rank–nullity theorem

Definition 4.5.10. The **rank** of f is $\dim(\mathrm{im} f)$; its **nullity**, $\dim(\ker f)$. *(Consistency: for $f(x) = Ax$,* $\mathrm{im} f$ *is the span of the columns, so the rank of f is the rank of A.)*

Theorem 4.5.11 (rank–nullity). If $f : V \to W$ is linear and $\dim V = n$ (finite), then

$$\dim V = \dim(\ker f) + \dim(\mathrm{im} f).$$

Proof. The kernel, as a subspace of V, has a basis $\{u_1, \dots, u_k\}$ (with $k = \dim \ker f$), which can be extended to a basis $\{u_1, \dots, u_k, v_1, \dots, v_{n-k}\}$ of V — both facts by **Lemma 4.3.18b**. We claim that $\{f(v_1), \dots, f(v_{n-k})\}$ is a **basis of** $\mathrm{im} f$, which gives $\dim \mathrm{im} f = n - k$ and closes the theorem.

They generate: every element of the image is $f(v)$ with $v = \sum_i a_i u_i + \sum_j b_j v_j$; applying f and using $f(u_i) = 0$: $f(v) = \sum_j b_j f(v_j)$.

Independent: if $\sum_j b_j f(v_j) = 0$, by linearity $f(\sum_j b_j v_j) = 0$, so $\sum_j b_j v_j \in \ker f$ and it can be written $\sum_i a_i u_i$. Then $\sum_i a_i u_i - \sum_j b_j v_j = 0$ is a null combination of the basis of V: all coefficients are zero, in particular all the b_j. ∎

Corollary 4.5.12 (the matrix case is bookkeeping). For $f(x) = Ax$ with A of size $m \times n$: the nullity is $n - \operatorname{rank} A$ (Theorem 4.4.6) and the rank is $\operatorname{rank} A$, and indeed $n = (n - \operatorname{rank} A) + \operatorname{rank} A$. *(That is why in the matrix case "there was nothing to prove": both summands were already known via pivots. The content of Theorem 4.5.11 is the **general** case, with no matrices and no pivots.)*

Remark 4.5.12b (where each object lives: the classic mistake). The kernel and the image **do not live in the same space**: $\ker f \subseteq V$ (the source space) and $\operatorname{im} f \subseteq W$ (the target one). Theorem 4.5.11 does **not** say that the image is a piece of V that "survives", nor that $V = \ker f \oplus \operatorname{im} f$: that does not even make sense when $V \neq W$. The only thing that adds up is the **dimension**: that of V is split between what gets flattened (inside V) and what the image occupies (inside W). Exercise 3 is the warning turned into a computation: for $f(x, y, z) = (x + y,\ y + z)$, the kernel is a line in $\mathbb{R}^3$ and the image is **all** of $\mathbb{R}^2$, and $3 = 1 + 2$ without there being any sum to take inside $\mathbb{R}^3$.

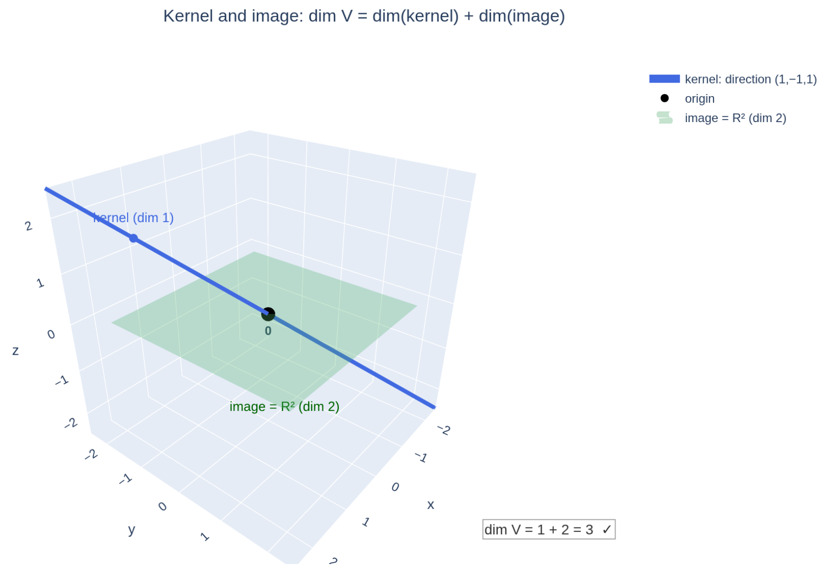

Figure 10: Kernel and image of an $f : \mathbb{R}^3 \to \mathbb{R}^2$: in blue, the kernel, a line **inside the source space** $\mathbb{R}^3$; in green, the image, which is **all** of $\mathbb{R}^2$ — the **target** space, drawn here as the plane $z = 0$ merely so that $1 + 2 = 3$ can be read off a single picture. The image is **not** a piece of $\mathbb{R}^3$: what adds up are the dimensions, not the subspaces (Remark 4.5.12b).

[see interactive version]

5.5 Isomorphisms

Definition 4.5.13. An **isomorphism** is a **bijective** linear map $f : V \to W$. If one exists, we say that V and W are **isomorphic** ($V \cong W$): as far as structure goes, the same space.

Lemma 4.5.14 (the inverse of a bijective linear map is linear). If $f : V \to W$ is linear and bijective, then $f^{-1} : W \to V$ is linear.

Proof. Let $w, w' \in W$ and $a, b \in K$. Since f is linear,

$$f(a\,f^{-1}(w) + b\,f^{-1}(w')) = a\,f(f^{-1}(w)) + b\,f(f^{-1}(w')) = aw + bw'.$$

Applying f^{-1} to both ends: $a\,f^{-1}(w) + b\,f^{-1}(w') = f^{-1}(aw + bw')$. ■

> *This was not free:* bijectivity gives an inverse *as a map of sets*; that it respects the structure is this lemma. **Consequence:** f^{-1} has a matrix, and from $f^{-1} \circ f = \mathrm{id}$ together with Proposition 4.5.6 we get (matrix of f^{-1}) $\cdot\, A = I$: the matrix of f^{-1} is exactly A^{-1} (Corollary 4.3.24, “one side suffices”). The inverse matrix of Session 2 is thus justified *as a map*.

Proposition 4.5.15 (coordinates, now with a name). If $\dim V = n$, the coordinate map $V \to K^n$ (once a basis is fixed) is an **isomorphism**. Therefore, **every n-dimensional space over K is isomorphic to K^n**.

Proof. It is linear (coordinates respect the operations, Session 3, ex. 6), injective (coordinates are unique, Proposition 4.3.18) and surjective (every tuple defines a vector). ■

Theorem 4.5.16 (characterization, closing the thread). For a square $n \times n$ matrix A, the following are equivalent: A invertible $\iff$ $x \mapsto Ax$ is an isomorphism $\iff$ $\operatorname{rank} A = n$.

Proof. If A is invertible, $x \mapsto A^{-1}x$ is an inverse of $x \mapsto Ax$ (both compositions give the identity by Theorem 4.2.5), so the latter is bijective: an isomorphism. If $x \mapsto Ax$ is an isomorphism, its inverse is linear (Lemma 4.5.14) with a matrix B such that $BA = I$, and by Corollary 4.3.24, A is invertible. The equivalence with $\operatorname{rank} A = n$ is Corollary 4.3.24 itself. ■

Proposition 4.5.17 (isomorphisms preserve dimension). If $V \cong W$ (finite dimensions), then $\dim V = \dim W$. In particular, there is **no** isomorphism between spaces of different dimensions.

Proof. If f is an isomorphism, $\ker f = \{0\}$ (injective: $f(v) = 0 = f(0) \Rightarrow v = 0$) and $\operatorname{im} f = W$ (surjective). By rank–nullity: $\dim V = 0 + \dim W$. ■

Exercises

1. ★ Decide, with proof, whether the following are linear: (a) $f(x, y) = (2x - y,\ x + 3y)$; (b) $g(x, y) = (x + 1,\ y)$; (c) $T(p) = x \cdot p'(x)$ on $K[x]_{\leq n}$; (d) $h(x, y) = |x| + |y|$.
2. ★ Find the matrix of the rotation by an angle θ in $\mathbb{R}^2$ (standard basis): compute where e_1 and e_2 go and place them as columns.
3. ★ For $f(x, y, z) = (x + y,\ y + z)$ from $\mathbb{R}^3$ to $\mathbb{R}^2$: find its matrix, compute $\ker f$ and $\operatorname{im} f$ with bases, and verify rank–nullity.
4. ★ **(Isomorphism?)** Decide, justifying with results from the session: (a) $f(x, y) = (x + y,\ x - y)$ on $\mathbb{R}^2$; (b) the derivative D on $K[x]_{\leq n}$; (c) some isomorphism between $\mathbb{R}^4$ and the 2×2 matrices (construct it); (d) can there be an isomorphism $\mathbb{R}^3 \to \mathbb{R}^2$?
5. Find the matrix of the derivative $D : K[x]_{\leq 2} \to K[x]_{\leq 2}$ in the basis $\{1, x, x^2\}$, and use it to recover $D(3 + 2x - x^2)$ by multiplying.
6. Prove that a linear f is injective $\iff$ $\ker f = \{0\}$. *(One direction is in 5.17; complete the other: if the kernel is trivial and $f(u) = f(v)$…)*
7. Write out in full detail the basis extension used in Theorem 4.5.11 for $f(x, y, z) = (x + y, y + z)$ of exercise 3.
8. Let $f : U \to V$, $g : V \to W$ be isomorphisms. Prove that $g \circ f$ is an isomorphism and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Going further and references

Going further (the quotient and the first isomorphism theorem). The rank–nullity theorem has an elegant structural version: constructing the **quotient space** $V/\ker f$ (identify vectors that differ by something in the kernel), one obtains $V/\ker f \cong \mathrm{im} f$ — the **first isomorphism theorem**. The construction of the quotient requires a careful development that exceeds this course; the interested reader can find it in Aluffi, *Algebra: Chapter 0*, ch. III and VI, or in Axler, ch. 3.E.

References. Linear maps and matrices: Axler, ch. 3.A–3.C; Hefferon, Three.I–III. Rank–nullity (there called the *Fundamental Theorem of Linear Maps*): Axler, 3.B. Isomorphisms: Axler, 3.D; Hefferon, Three.I.

Lecture Notes · Linear Algebra · Session 6 — Determinants

The closing of the module: the determinant defined by what it **measures** (oriented volume), the formula **derived** from three properties (with the uniqueness reasoned out), the computation by Gaussian elimination justified with elementary matrices, the central theorems proved ($\det AB$, invertibility, $\det A^\top$) and Cramer's rule with its why. Exercises and a Going further section at the end.

6.1 What the determinant measures

Scope warning. This section is the **geometric motivation**, and it lives **over** $\mathbb{R}$: "volume", "orientation" and "sign" make direct sense in $\mathbb{R}^n$ (and, with caveats, in ordered fields), but **not** in an arbitrary field — what would "volume" be in $\mathbb{Z}/5\mathbb{Z}$? The **algebraic** definition of the determinant, valid **over any field** K, is the one in §6.2 (the three properties); the geometry of this section is its reading when $K = \mathbb{R}$.

And a warning about logical status. That every region sees its volume multiplied by $|\det A|$ —and, all the more, that the function defined in §6.2 by its three properties **is** exactly that volume factor— **is not proved in this course**: doing it rigorously requires measuring volumes for real (measure theory, and the Jacobian of the change of variables; see the Going further section). Nothing in §6.2–§6.5 relies on it: those proofs are purely algebraic. Volume is here the **compass** telling us what the results mean, not a premise from which any of them is deduced.

Over $\mathbb{R}$: a transformation $x \mapsto Ax$ of $\mathbb{R}^n$ deforms space. The determinant captures, in a single number, **the factor by which every volume is multiplied**: every region sees its n-dimensional volume multiplied by $|\det A|$. The **unit n-cube** is just the normalized witness: since its volume is 1, its image (an **n-parallelepiped**, the one generated by the columns of A) has volume exactly $|\det A|$. In the plane: the unit square goes to a parallelogram of area $|\det A|$.

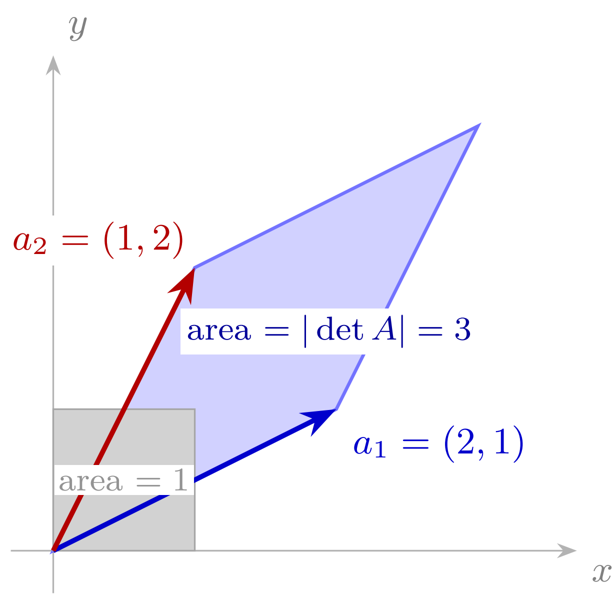

Figure 11: The determinant as an area factor: the matrix A takes the unit square (area 1) to a parallelogram of area $|\det A|$.

The sign: orientation. The determinant carries a sign, and the sign says whether the transformation **preserves or reverses the orientation** of space. In the plane, orientation is the *sense of rotation* (from e_1 towards e_2, counterclockwise): a transformation preserving it has $\det > 0$; a **reflection** (mirror) reverses it: $\det < 0$. In space it is the phenomenon of the right hand turned into a left hand. And the degenerate case:

$\det A = 0 \iff$ the transformation **flattens** space into something of lower dimension,

an equivalence we shall prove (Theorem 4.6.8) in the form "$\det A = 0 \iff A$ not invertible".

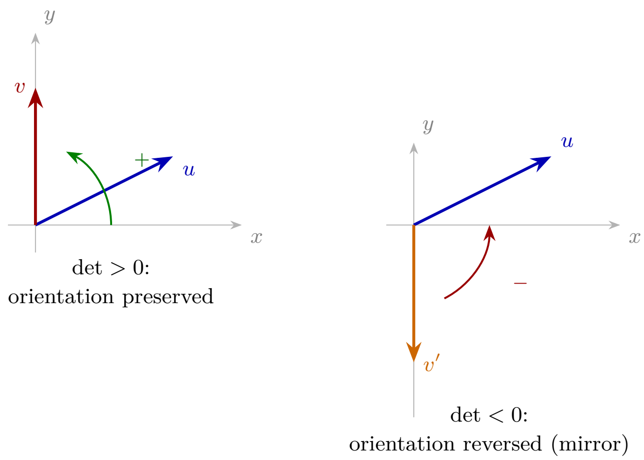

Figure 12: The sign of the determinant is the orientation: det > 0 preserves it, det < 0 reverses it (mirror reflection).

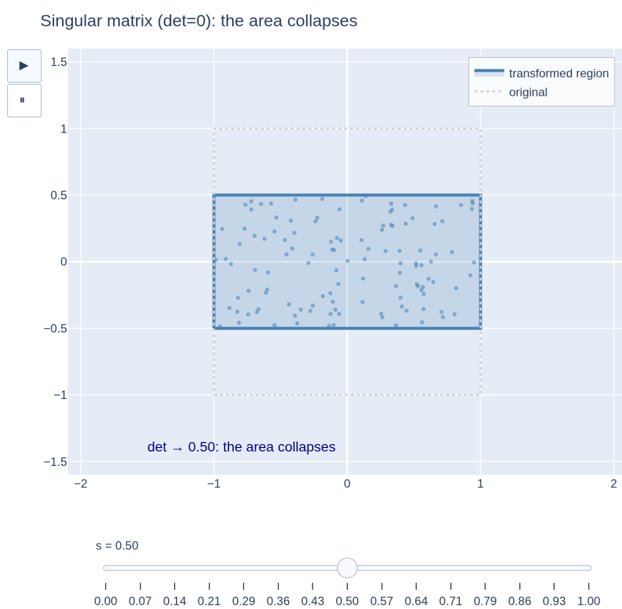

Figure 13: As det $A \to 0$ the transformation flattens the area to zero: a singular matrix collapses space to a lower dimension.

[see interactive version]

6.2 The axiomatic definition

Instead of giving a formula, we impose the properties that an oriented volume **must** have.

Definition 4.6.1. A **determinant function** is a map det assigning to each $n \times n$ matrix over K a scalar, satisfying:

- **(D1) Multilinear in the rows:** with the other rows fixed, det is "linear in each row": if row i is $u + v$, $\det = \det(\dots u \dots) + \det(\dots v \dots)$; and if it is λu, $\det = \lambda \det(\dots u \dots)$. *(Stretching one side scales the volume; splitting one side splits the volume.)*
- **(D2) Alternating:** if two rows are equal, $\det = 0$. *(A flattened body has no volume.)*
- **(D3) Normalization:** $\det I = 1$. *(The unit cube has volume 1.)*

  Notation (what those dots mean). In expressions such as $\det(\dots, u, \dots, v, \dots)$ the dots are **not** an "and so on": they are **fixed rows**. Row i is u, row j is v, and all the others are **the same in every term** of the identity — the only thing that changes from one term to another is what is written out. That is precisely why the terms can be compared and cancelled against each other, which is what the proofs below do.

  Notation (the dots). In expressions such as $\det(\dots, u, \dots, v, \dots)$ the dots are **not** an "and so on": they are **fixed rows**. The notation means *row i equals u, row j equals v, and all the other rows are whatever they are* — **the same ones in every term** of the identity where they appear. That is exactly what lets the terms be compared and cancelled against each other: the only thing that changes from one to another is what is written out.

Proposition 4.6.2 (immediate consequences). For a determinant function: 1. A zero row $\Rightarrow \det = 0$. 2. Swapping two rows changes the sign. 3. Adding to a row a multiple of another **does not change** det. 4. Multiplying **one** row by λ multiplies det by λ; consequently, $\det(\lambda A) = \lambda^n \det A$ *(all n rows get scaled: the n-cube of side λ has volume λ^n — not to be confused with scaling a single row)*. 5. Linearly **dependent** rows $\Rightarrow \det = 0$.

Proof. (1) A zero row $= 0 \cdot$ (zero row); by (D1), $\det = 0 \cdot \det = 0$. (2) Let u, v be rows i, j. By (D2), the matrix with $u + v$ in **both** positions has $\det = 0$; expanding by (D1) in the two rows: $0 = \det(\dots u \dots u \dots) + \det(\dots u \dots v \dots) + \det(\dots v \dots u \dots) + \det(\dots v \dots v \dots)$. The first and the last vanish by (D2), hence $\det(\dots u \dots v \dots) = -\det(\dots v \dots u \dots)$. (3) $\det(\dots u + \mu w \dots w \dots) = \det(\dots u \dots w \dots) + \mu \det(\dots w \dots w \dots) = \det(\dots u \dots w \dots) + 0$. (4) Direct from (D1); iterating over the n rows, λ^n. (5) If a row is a combination of the others, subtracting that combination from it (operations of type (3), which do not change det) leaves a zero row: $\det = 0$ by (1). ∎

Properties (2)–(4) are **exactly** the effects of the three elementary operations: the direct derivation from the axioms, cumbersome but elementary, which the video refers here. The clean route via elementary matrices comes in §6.3.

Example 4.6.3 (the 2×2 case, forced). Let det be a determinant function on 2×2 matrices. Writing the rows in the standard basis, $(a, b) = ae_1 + be_2$ and $(c, d) = ce_1 + de_2$, and expanding by (D1) twice:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ac \det \begin{pmatrix} e_1 \\ e_1 \end{pmatrix} + ad \det \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} +$$

$$+ bc \det \begin{pmatrix} e_2 \\ e_1 \end{pmatrix} + bd \det \begin{pmatrix} e_2 \\ e_2 \end{pmatrix} = ad - bc,$$

using (D2) to kill the first and the last, (D3) for $\det(e_1, e_2) = 1$ and Proposition 4.6.2.2 for $\det(e_2, e_1) = -1$. **There was no choice:** the formula is forced.

Theorem 4.6.4 (existence and uniqueness). For each n there exists **exactly one** determinant function on $n \times n$ matrices.

Idea of the proof (uniqueness). The argument of Example 4.6.3 scales: expanding each row in the standard basis by (D1), $\det A$ becomes a sum of terms $a_{1j_1} a_{2j_2} \cdots a_{nj_n} \det(e_{j_1}, \dots, e_{j_n})$. If some index repeats, (D2) kills the term; only the **rearrangements** of $(1, \dots, n)$ remain, and each $\det(e_{j_1}, \dots, e_{j_n})$ equals ± 1: the sign resulting from counting the swaps that sort the list (4.6.2.2 and D3). The value of $\det A$ is thus completely determined — this is the **Leibniz formula** (Going further). *Existence:* the formula itself defines a function verifying (D1)–(D3); the check, routine but long, can be found in Hefferon, Four.I. ■

6.3 Computation

Proposition 4.6.5 (determinant of a triangular matrix). If A is triangular (upper or lower), $\det A = a_{11} a_{22} \cdots a_{nn}$ (product of the diagonal).

Proof. If all $a_{ii} \neq 0$: with operations of type "add a multiple of another row" (which do not change det, 4.6.2.3) the off-diagonal elements are killed (working from the last row upwards in the upper triangular case); a diagonal matrix remains, and by (D1) applied row by row, $\det = a_{11} \cdots a_{nn} \det I = a_{11} \cdots a_{nn}$. If some $a_{ii} = 0$: the product is 0, and also $\det A = 0$, since columns $1, \dots, i$ live in the span of $e_1, \dots, e_{i-1}$ (in the upper case), hence they are dependent, the rank is $< n$, the **rows** are dependent (Theorem 4.3.23) and 4.6.2.5 applies. ■

Proposition 4.6.5b (determinant by triangular blocks). If M has the form

$$M = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$$

with A and D **square** (of sizes k and m) and a block of zeros below A, then $\det M = \det A \cdot \det D$. In particular, for diagonal blocks ($B = 0$ as well).

Proof. Reduce A operating only on the first k rows of M: those operations do not touch the zero block (adding and swapping rows that start with zeros leaves zeros), and their accumulated effect multiplies the determinant by the same factor $c_1 \neq 0$ both in M and in A (Proposition 4.6.6, iterated). Then reduce D with the last m rows (factor c_2). The result M' is **triangular**, its diagonal being that of A' followed by that of D'; by Proposition 4.6.5,

$$c_1 c_2 \det M = \det M' = (\text{diag } A')(\text{diag } D') = \det A' \cdot \det D' = c_1 \det A \cdot c_2 \det D,$$

and cancelling $c_1 c_2$, $\det M = \det A \cdot \det D$. ■

> **Warning:** the shortcut requires the block of **zeros**. For a general $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ there is **no** naive formula in terms of the four block determinants (exercise 10: a counterexample).

> **Example 4.6.5c.** $\det \begin{pmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 1 & 3 \end{pmatrix} = \det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \cdot \det \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = (-2)(1) = -2$
> — two 2×2 determinants instead of one 4×4.

Proposition 4.6.6 (determinants of the elementary matrices). $\det E_{(\mathrm{T1})} = -1$; $\det E_{(\mathrm{T2}),\lambda} = \lambda$; $\det E_{(\mathrm{T3}),\mu} = 1$. Moreover, for any M:

$$\det(EM) = \det(E)\ \det(M).$$

Proof. The values: each E is the identity after one operation, and (4.6.2.2–4.6.2.4) give its effect on $\det I = 1$. The equality: EM is M after the operation (Lemma 4.2.10), and the effect of the operation on $\det M$ (4.6.2.2–4.6.2.4) is to multiply by -1, λ or 1 — exactly $\det E$. ■

Method (computation by Gaussian elimination). Reduce A to echelon form keeping track of the effects: each swap contributes a factor -1, each scaling a λ, the additions of multiples nothing; upon reaching a triangular matrix, multiply the diagonal (4.6.5) and adjust the factors. For 3×3 there is also the **rule of Sarrus**, and in general the *cofactor expansion* (reference: Hefferon, Four.III); for large sizes, Gaussian elimination is the efficient route.

6.4 The central theorems

Theorem 4.6.7 ($\det$ **of a product).** $\det(AB) = \det(A)\ \det(B)$.

Proof. Case A invertible: by Corollary 4.3.24, $A = E_1 E_2 \cdots E_k$ is a product of elementary matrices. Applying Proposition 4.6.6 repeatedly:

$$\det(AB) = \det(E_1 \cdots E_k B) = \det(E_1) \cdots \det(E_k)\ \det(B),$$

and taking $B = I$ in this same chain, $\det(A) = \det(E_1) \cdots \det(E_k)$. Combining, $\det(AB) = \det(A) \det(B)$. *Case A not invertible:* then $\operatorname{rank} A < n$, and the rows of A are dependent (Theorem 4.3.23), so $\det A = 0$ (4.6.2.5). On the other hand, the columns of AB are linear combinations of the columns of A (each one is $A \cdot$ (column of B), Proposition 4.2.3), so $\operatorname{rank}(AB) \leq \operatorname{rank} A < n$; the rows of AB are also dependent (Theorem 4.3.23 again) and $\det(AB) = 0$ (4.6.2.5). Both sides equal 0 — with no need to invoke Theorem 4.6.8. ■

Theorem 4.6.8 (invertibility). A is invertible $\iff$ $\det A \neq 0$. And if it is, $\det(A^{-1}) = 1/\det A$.

Proof. Reduce A to a triangular T. Each operation multiplies the determinant by a known and **nonzero** factor (-1 for a swap, $\lambda \neq 0$ for a scaling, 1 when adding a multiple of another row, Proposition 4.6.2), so that

$$\det T = c \cdot \det A, \qquad c = \text{product of those factors} \neq 0$$

—the same direction as in Proposition 4.6.5b— and, dividing by c, $\det A = c^{-1} \det T$. In particular $\det A$ and $\det T$ **vanish together**. If A is invertible, its rank is n (Corollary 4.3.24), T has n pivots —a diagonal with no zeros— and $\det T \neq 0$ (4.6.5), so $\det A \neq 0$. If A is not invertible, its rows are dependent (rank $< n$, Theorem 4.3.23) and $\det A = 0$ (4.6.2.5). The formula for the inverse: $1 = \det I = \det(AA^{-1}) = \det A \cdot \det(A^{-1})$ by 4.6.7. ■

Proposition 4.6.9 (transpose). $\det(A^{\mathsf{T}}) = \det A$.

Proof. Invertible case: $A = E_1 \cdots E_k$, hence $A^{\mathsf{T}} = E_k^{\mathsf{T}} \cdots E_1^{\mathsf{T}}$ (Session 2, ex. 6). It suffices to see $\det(E^{\mathsf{T}}) = \det(E)$ for each type: the swap and scaling ones are **symmetric** ($E^{\mathsf{T}} = E$), and the transpose of one of type (T3) is another of type (T3) (the μ changes position), all with det $= 1$. With 6.7: $\det(A^{\mathsf{T}}) = \det(E_k) \cdots \det(E_1) = \det A$. *Non-invertible case:* $\operatorname{rank} A^{\mathsf{T}} = \operatorname{rank} A < n$ (Theorem 4.3.23: transposing interchanges rows and columns), and both determinants are 0. ■

> **Useful consequence:** everything said for rows holds for **columns** (multilinearity, alternation, effects of column operations): just transpose. We use it right away.

Theorem 4.6.10 (Cramer's rule). Let A be invertible ($n \times n$) and $Ax = b$. Then the (unique) solution is given by

$$x_i = \frac{\det A_i(b)}{\det A}, \qquad i = 1, \dots, n,$$

where $A_i(b)$ is A with **column** i **replaced by** b.

Proof. By the reading by columns, $b = Ax = x_1 c_1 + \cdots + x_n c_n$. Then, by multilinearity **in columns** (legitimate by 4.6.9):

$$\det A_i(b) = \det A_i\Big(\textstyle\sum_j x_j c_j\Big) = \sum_j x_j \det A_i(c_j).$$

For $j \neq i$, the matrix $A_i(c_j)$ has column j **repeated** in position i: determinant 0 (alternation in columns). Only $j = i$ remains: $\det A_i(c_i) = \det A$. Hence

$\det A_i(b) = x_i \det A$, and x_i is isolated **dividing by** $\det A \neq 0$ — the field, one last time. ■

> Every unknown is a **quotient of volumes**. Elegant, but inefficient for computing: in practice, Gaussian elimination.

6.5 The map of the module

Theorem 4.6.11 (complete characterization). For a square $n \times n$ matrix A, the following are equivalent:

1. A is invertible;
2. $\det A \neq 0$;
3. $\operatorname{rank} A = n$;
4. $x \mapsto Ax$ is an isomorphism;
5. the columns (or rows) of A are linearly independent;
6. $Ax = 0$ has only the trivial solution.

(Proof: Corollary 4.3.24 + Theorems 4.5.16 and 4.6.8.) Geometrically: to invert is **to be able to undo**; with $\det \neq 0$ volumes are multiplied by a nonzero factor —nothing collapses, and one undoes by dividing by it ($\det A^{-1} = 1/\det A$)—; with $\det = 0$ the volume collapses and **there is no way back**.

Software (RA7) — the same system in several worlds. Here the whole module comes together: the determinant, the solving of the system, and what happens when the field of coefficients changes. The call `A.inv_mod(5)` **fails on purpose**: before running it, compute $\det A$ modulo 5 and predict the error. It is Theorem 4.6.11 checking itself.

```python
# The same system in several worlds (det A = 10)
from sympy import Matrix
A = Matrix([[2, 1, -1], [1, 3, 2], [1, 0, 1]])
b = Matrix([8, 13, 4])
print("det A =", A.det())
print("invertible? (Theorem 4.6.11):", A.det() != 0)
print("over Q:", A.solve(b).T)
print("mod 3 (a field, nonzero det):", A.inv_mod(3).tolist())
try:
    A.inv_mod(5)
except Exception as e:
    print("mod 5: FAILS -", e)
# Prediction: what about mod 7? and mod 6? Try it.
```

Those three calls sum up the whole session: `A.det()` gives the determinant (**exact**, with no rounding); `A.det() != 0` answers the characterization of Theorem 4.6.11 —and by Theorem 4.6.8 it is the same information as the previous number—; and `A.solve(b)` solves $Ax = b$. If you work in **SageMath** instead of sympy, the same three things are written `A.det()`, `A.is_invertible()` and `A.solve_right(b)`. With **NumPy** they would be `np.linalg.det(A)` and `np.linalg.solve(A, b)`, but in floating point: the exact 0 becomes a tiny number and the criterion "$\det A \neq 0$" stops being reliable — one more reason to stay here in exact arithmetic. What no environment does for you is knowing **what those numbers mean**: why the determinant decides invertibility, and what has happened to the volume.

Guided activity (RA7-1). This box has an activity of its own, with the prediction table to fill in *before* running anything and the $\mathbb{Z}/6\mathbb{Z}$ case, where the determinant is non-zero and yet there is no inverse. The **full statement** is in Activity RA7-1 · The determinant decides, and it changes with the field. Submission goes through the virtual classroom.

Exercises

1. ★ Compute, by the method of your choice, justifying the steps: $\det\begin{pmatrix}1&2&3\\4&5&6\\7&8&10\end{pmatrix}$ and $\det\begin{pmatrix}2&0&0\\5&3&0\\1&4&-1\end{pmatrix}$.
2. ★ Decide whether $\begin{pmatrix}1&1&2\\1&2&3\\2&3&5\end{pmatrix}$ is invertible using only its determinant (compute it by Gaussian elimination).
3. ★ Check with a pair of concrete 2×2 matrices: $\det(AB) = \det A\cdot\det B$ and $\det(A+B) \neq \det A + \det B$ (the determinant is **not** additive in the whole matrix — only multilinear by rows!).
4. ★ Verify $\det(\lambda A) = \lambda^2 \det A$ with a 2×2 matrix, and explain the exponent geometrically. What is $\det(2A)$ if A is 5×5 with $\det A = 3$?
5. Prove from the axioms that a matrix with two equal columns has determinant 0. *(Hint: 4.6.9.)*
6. Solve by Cramer the system $\{2x + y = 3;\ x - y = 0\}$ and check with Gaussian elimination.
7. Interpret: if $\det A = -2$ in $\mathbb{R}^2$, what happens to areas and to orientation under $x \mapsto Ax$?
8. Deduce from Theorem 4.6.7 that $\det(A^k) = (\det A)^k$, and that if $A^k = I$ for some $k \geq 1$, then $\det A$ is a k-th root of unity in K (a connection with the Complex Numbers module when $K = \mathbb{C}$!).
9. True or false? "If $\det A = \det B$, then A and B have the same rank." Justify or give a counterexample.
10. **(The warning of 4.6.5b, fulfilled.)** Let M be the 4×4 matrix with

2×2 blocks $A = D = 0$ and $B = C = I$. Check that $\det M = 1$ *(hint: M permutes $e_1 \leftrightarrow e_3$ and $e_2 \leftrightarrow e_4$: two swaps)*, and conclude that for general blocks **no** naive formula like $\det M = \det A \cdot \det D - \det B \cdot \det C$ can hold (here it would give -1).

Going further and references

Going further (the Leibniz formula). Uniqueness (Theorem 4.6.4) produces the explicit formula

$$\det A = \sum_{\sigma} \operatorname{sgn}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)},$$

a sum over all **permutations** σ of $\{1, \dots, n\}$, with $\operatorname{sgn}(\sigma) = \pm 1$ according to the parity of swaps. It has $n!$ terms —which is why one does not compute with it— but it is the theoretical reference: from it come, for instance, the cofactor expansion and another proof of $\det A^{\mathsf{T}} = \det A$. See Hefferon, Four.I, or Aluffi, ch. VIII.

Going further (volume and integration). The determinant as a volume factor is exactly the **Jacobian** of the change of variables in multiple integrals — the bridge to the calculus you will study later on.

References. Axiomatics and uniqueness: Hefferon, Four.I; Strang, ch. 5. Cramer: Hefferon, Four.III; de Burgos. Orientation and volume: Strang, §5.3.